

Stuff I've derived

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A Numerical Methods

A.1 Finite Difference Stencils

A.1.1 First Derivatives

Two-point forward difference stencil with error:

$$f'(x) = \frac{f(x + \varepsilon) - f(x)}{\varepsilon} - \frac{\varepsilon}{2} f''(x) + \mathcal{O}(\varepsilon^2) \quad (\text{A.1.1})$$

Three-point central difference stencil with error:

$$f'(x) = \frac{f(x + \varepsilon) - f(x - \varepsilon)}{2\varepsilon} - \frac{\varepsilon^2}{6} f^{(3)}(x) + \mathcal{O}(\varepsilon^3) \quad (\text{A.1.2})$$

Five-point central difference stencil with error:

$$f'(x) = \frac{-f(x + 2\varepsilon) + 8f(x + \varepsilon) - 8f(x - \varepsilon) + f(x - 2\varepsilon)}{12\varepsilon} + \frac{\varepsilon^4}{30} f^{(5)}(x) + \mathcal{O}(\varepsilon^6) \quad (\text{A.1.3})$$

A.1.2 Second Derivatives

A.1.2.1 Single-Variable

Three-point stencil with error:

$$f''(x) = \frac{f(x + \varepsilon) - 2f(x) + f(x - \varepsilon)}{\varepsilon^2} - \frac{\varepsilon^2}{12} f^{(4)}(x) + \mathcal{O}(\varepsilon^4) \quad (\text{A.1.4})$$

Five-point stencil with error:

$$f''(x) = \frac{-f(x + 2\varepsilon) + 16f(x + \varepsilon) - 30f(x) + 16f(x - \varepsilon) - f(x - 2\varepsilon)}{12\varepsilon^2} - \frac{\varepsilon^4}{90} f^{(6)}(x) + \mathcal{O}(\varepsilon^6) \quad (\text{A.1.5})$$

A.1.2.2 Mixed Second Derivative

With a grid spacing of $\varepsilon_x, \varepsilon_y > 0$ in the x - and y -directions, respectively, we have

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{f_{++} + f_{--} - f_{+-} - f_{-+}}{4\varepsilon_x \varepsilon_y} - \frac{1}{6} \left(\varepsilon_x^2 \frac{\partial^4 f}{\partial x^3 \partial y} + \varepsilon_y^2 \frac{\partial^4 f}{\partial x \partial y^3} \right) + \mathcal{O}(\varepsilon^4) \quad (\text{A.1.6})$$

Here,

$$f_{\pm_1 \pm_2} = f(x \pm_1 \varepsilon_x, y \pm_2 \varepsilon_y) \quad (\text{A.1.7})$$

Note that the expression (A.1.6) corresponds to taking twice the central difference (A.1.2) of f , once with respect to x and once with respect to y .

A.1.3 Laplacian

27-point stencil:

$$\Delta f = \frac{1}{\varepsilon^2}((-8\lambda - 6) \cdot (\text{center}) + (4\lambda + 1) \cdot (\text{faces}) - 2\lambda \cdot (\text{edges}) + \lambda \cdot (\text{corners})) \quad (\text{A.1.8})$$

where, using $f_{+0-} = f(x + \varepsilon, y, z - \varepsilon)$ etc.,

$$\begin{aligned} (\text{center}) &= f_{000} \\ (\text{faces}) &= f_{+00} + f_{-00} + f_{0+0} + f_{0-0} + f_{00+} + f_{00-} \\ (\text{edges}) &= f_{++0} + f_{+-0} + f_{-+0} + f_{--0} \\ &\quad f_{+0+} + f_{+0-} + f_{-0+} + f_{-0-} \\ &\quad f_{0++} + f_{0+-} + f_{0-+} + f_{0--} \\ (\text{corners}) &= f_{+++} + f_{++-} + f_{+-+} + f_{-++} \\ &\quad f_{+--} + f_{-+-} + f_{--+} + f_{---}. \end{aligned} \quad (\text{A.1.9})$$

The error is $\mathcal{O}(\varepsilon^2)$, with leading contribution given by

$$\frac{\varepsilon^2}{12} \left(\frac{\partial^4 f}{\partial x^4} + \frac{\partial^4 f}{\partial y^4} + \frac{\partial^4 f}{\partial z^4} \right). \quad (\text{A.1.10})$$

The choice of λ doesn't change this leading error order, but it changes the anisotropy of the error. Useful choices are $\lambda = \frac{1}{22}$ or $\lambda = \frac{1}{26}$. The latter minimises the anisotropy for Fourier modes, and its full expression reads

$$\Delta f = \frac{-164 \cdot (\text{center}) + 30 \cdot (\text{faces}) - 2 \cdot (\text{edges}) + 1 \cdot (\text{corners})}{26\varepsilon^2} + \mathcal{O}(\varepsilon^2)_{\text{iso}} + \mathcal{O}(\varepsilon^4)_{\text{aniso}}. \quad (\text{A.1.11})$$

A.2 Runge-Kutta

A.2.1 General Structure

Runge-Kutta is the name of a family of integrators for first-order differential equations of the form

$$y'(t) = F(t, y(t)). \quad (\text{A.2.1})$$

Generally speaking, given a timestep ε , one evaluates the right-hand side of the ODE s times, at different times $t_i = \varepsilon c_i \in (t_0, t_0 + \varepsilon)$ and for different guesses for y . Concretely, starting from $y(t_0) = y_0$, one evaluates

$$\begin{aligned} k_1 &= F(t_0, y_0), \\ k_2 &= F(t_0 + \varepsilon c_2, y_0 + \varepsilon a_{21} k_1), \\ k_3 &= F(t_0 + \varepsilon c_3, y_0 + \varepsilon(a_{32} k_2 + a_{31} k_1)), \\ &\vdots \\ k_s &= F(t_0 + \varepsilon c_s, y_0 + \varepsilon(a_{s,s-1} k_{s-1} + \dots + a_{s1} k_1)). \end{aligned} \quad (\text{A.2.2})$$

These are then used to approximate $y(t + \varepsilon)$ as

$$y(t + \varepsilon) \approx y_0 + \varepsilon(b_1 k_1 + \dots + b_s k_s). \quad (\text{A.2.3})$$

An RK integrator is hence defined by picking a value of s , quadrature points $c_i \in [0, 1]$, $i = 2, \dots, s$, as well as the coefficients b_i , $i = 1, \dots, s$ and a_{ij} , $i > j$. These coefficients are not chosen at random, but rather picked such that the error of $y(t + \varepsilon)$ is as small as possible, i.e. of as high of an order ε as can be arranged.

The coefficients can be thought of as a matrix $\mathbf{A} \in \mathbb{R}^{s \times s}$ and a pair of vectors $\mathbf{b}, \mathbf{c} \in \mathbb{R}^s$, with the coefficients arranged such that

$$\mathbf{A} = \begin{pmatrix} 0 & & & & 0 \\ a_{21} & 0 & & & 0 \\ a_{31} & a_{32} & 0 & & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ a_{s1} & a_{s2} & \dots & a_{s,s-1} & 0 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_s \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_s \end{pmatrix}. \quad (\text{A.2.4})$$

Further writing $\mathbf{k} = (k_1, \dots, k_s)^\top$, and denoting row evaluation of a vector matrix by $[\cdot]_i$, we can write the above as follows:

$$k_i = F(t_0 + \varepsilon[\mathbf{c}]_i, y_0 + \varepsilon[\mathbf{A}\mathbf{k}]_i), \quad y(t_0 + \varepsilon) \approx y_0 + \varepsilon \mathbf{b}^\top \mathbf{k}. \quad (\text{A.2.5})$$

One way of determining optimal coefficients $\mathbf{A}, \mathbf{b}, \mathbf{c}$ is to expand the approximation (A.2.3) around $\varepsilon = 0$ and comparing coefficients to the Taylor expansion of $y(t_0 + \varepsilon)$, which using $y' = F(t, y(t))$ as well as the notation

$$F = F(t_0, y_0), \quad F_t = (\partial_t F)(t_0, y_0), \quad F_y = (\partial_y F)(t_0, y_0) \quad \text{etc.} \quad (\text{A.2.6})$$

can be written explicitly as

$$\begin{aligned} y(t_0 + \varepsilon) &= y_0 + \varepsilon F + \frac{\varepsilon^2}{2} (F_t + F F_y) \\ &\quad + \frac{\varepsilon^3}{6} (F_{tt} + 2F F_{ty} + F_t F_y + F F_y^2 + F^2 F_{yy}) \\ &\quad + \frac{\varepsilon^4}{24} (F_{ttt} + 3F F_{tty} + 3F_t F_{ty} + 5F F_y F_{ty} + 3F^2 F_{tyy} + F_{tt} F_y \\ &\quad \quad + 3F F_t F_{yy} + F_t F_y^2 + F F_y^3 + 4F^2 F_y F_{yy} + F^3 F_{yyy}) \\ &\quad + \mathcal{O}(\varepsilon^5) \end{aligned} \quad (\text{A.2.7})$$

As is evident from the number of terms quickly growing with the order in ε , the strategy of comparing coefficients quickly becomes unmanageable for higher-order RK integrators. Further, we can see that it is highly non-linear in derivatives of F ; this explains why we need to use the k_i with $i < j$ in the evaluation of k_j . Without this chaining, we could not reproduce the nonlinearity in F required to match the explicit expression (A.2.7).

A.2.2 RK1 (is just Euler)

As a first illustration of how the strategy of reading off coefficients works, let us indulge in the lowest-order case of $s = 1$. Although *technically*, $s = 0$ would be the lowest-order case, and $y(t) \approx y_0$ is *technically* an approximation of the solution to $y' = F(t, y)$, it is neither a good nor interesting one, hence the decision to start at $s = 1$. We start by writing down the k 's, of which there is only one—namely

$$k_1 = F(t_0, y_0) = F. \quad (\text{A.2.8})$$

The RK method then prescribes the approximation

$$y(t_0 + \varepsilon) \approx y_0 + \varepsilon b_1 k_1 = y_0 + \varepsilon b_1 F \quad (\text{A.2.9})$$

For this to match expansion (A.2.7) up to order $\mathcal{O}(\varepsilon)$, we need to set $b_1 = 1$. Thus, RK1 is nothing but

$$y(t_0 + \varepsilon) = y_0 + \varepsilon F(t_0, y_0) + \mathcal{O}(\varepsilon^2), \quad (\text{A.2.10})$$

which we immediately recognise as the explicit Euler method.

A.2.3 RK2 (gets interesting)

Moving up to the $s = 2$ case, we can employ the same strategy to obtain a (hopefully) more precise integration scheme. Here, the k 's, along with their expansions in ε , read

$$\begin{aligned} k_1 &= F(t_0, y_0) = F, \\ k_2 &= F(t_0 + \varepsilon c_2, y_0 + \varepsilon a_{21} k_1) = F + \varepsilon c_2 F_t + \varepsilon a_{21} k_1 F_y + \mathcal{O}(\varepsilon^2) \\ &= F + \varepsilon (c_2 F_t + a_{21} F F_y) + \mathcal{O}(\varepsilon^2). \end{aligned} \quad (\text{A.2.11})$$

Again assembling the RK-approximation, we find

$$y(t_0 + \varepsilon) = y_0 + \varepsilon(b_1 + b_2)F + \varepsilon^2(b_2 c_2 F_t + b_2 a_{21} F F_y) + \mathcal{O}(\varepsilon^3) \quad (\text{A.2.12})$$

Clearly, $b_1 + b_2 = 1$ and $b_2 \neq 0$ must hold for this to have a chance of matching expansion (A.2.7) up to $\mathcal{O}(\varepsilon^2)$. Further, we need $b_2 c_2 = b_2 a_{21} = \frac{1}{2}$, which upon introduction of a parameter $\alpha \in [\frac{1}{2}, \infty)$ allows us to parameterise

$$b_1 = 1 - \alpha, \quad b_2 = \alpha, \quad c_2 = \frac{1}{2\alpha}, \quad a_{21} = \frac{1}{2\alpha}. \quad (\text{A.2.13})$$

Using this parametrisation, the RK2 integration procedure reads

$$k_1 = F(t_0, y_0), \quad k_2 = F\left(t_0 + \frac{\varepsilon}{2\alpha}, y_0 + \frac{\varepsilon}{2\alpha} k_1\right), \quad (\text{A.2.14})$$

and

$$y(t_0 + \varepsilon) = y_0 + \varepsilon((1 - \alpha)k_1 + \alpha k_2) + \mathcal{O}(\varepsilon^3). \quad (\text{A.2.15})$$

Two common choices for the free parameter α are

- $\alpha = \frac{1}{2}$ (Heun's method): This leads to the approximation scheme

$$y(t_0 + \varepsilon) = y_0 + \frac{\varepsilon}{2}(k_1 + k_2) + \mathcal{O}(\varepsilon^3) \quad (\text{A.2.16})$$

with

$$k_1 = F(t_0, y_0), \quad k_2 = F(t_0 + \varepsilon, y_0 + \varepsilon k_1). \quad (\text{A.2.17})$$

- $\alpha = 1$ (Explicit Midpoint Method): In this case, we get

$$y(t_0 + \varepsilon) = y_0 + \varepsilon k_2 + \mathcal{O}(\varepsilon^3), \quad (\text{A.2.18})$$

where

$$k_1 = F(t_0, y_0), \quad k_2 = F\left(t_0 + \frac{\varepsilon}{2}, y_0 + \frac{\varepsilon}{2} k_1\right). \quad (\text{A.2.19})$$

Although both these cases—and for that matter, any $\alpha \in [\frac{1}{2}, \infty)$ has the same *order* error, $\mathcal{O}(\varepsilon^3)$, different values for α lead to different forms of the error, which might—depending on the ODE in question—lead to a better constant factor.

A.2.4 RK3 (gets ugly)

For $s = 3$, the k 's and their expansions around $\varepsilon = 0$ read

$$\begin{aligned}
k_1 &= F(t_0, y_0) = F, \\
k_2 &= F(t_0 + \varepsilon c_2, y_0 + \varepsilon a_{21} k_1) \\
&= F + \varepsilon(c_2 F_t + a_{21} F F_y) + \frac{\varepsilon^2}{2}(c_2^2 F_{tt} + 2c_2 a_{21} F F_{ty} + a_{21}^2 F^2 F_{yy}) + \mathcal{O}(\varepsilon^3), \\
k_3 &= F(t_0 + \varepsilon c_3, y_0 + \varepsilon(a_{31} k_1 + a_{32} k_2)) \\
&= F + \varepsilon(c_3 F_t + (a_{31} + a_{32}) F F_y) + \frac{\varepsilon^2}{2}(2a_{32} c_2 F_t F_y + 2a_{32} a_{21} F F_y^2 \\
&\quad + c_3^2 F_{tt} + 2c_3(a_{31} + a_{32}) F F_{ty} + (a_{31} + a_{32})^2 F^2 F_{yy}) + \mathcal{O}(\varepsilon^3)
\end{aligned} \tag{A.2.20}$$

Next, we evaluate the RK3 integration scheme

$$y(t_0 + \varepsilon) \approx y_0 + \varepsilon(b_1 k_1 + b_2 k_2 + b_3 k_3) \tag{A.2.21}$$

order by order in ε , comparing to the explicit expansion (A.2.7) on the left-hand side:

$$\text{at } \mathcal{O}(\varepsilon): \quad F \stackrel{!}{=} b_1 F + b_2 F + b_3 F \tag{A.2.22}$$

which requires

$$b_1 + b_2 + b_3 = 1 \quad \Leftrightarrow \quad b_1 = 1 - b_2 - b_3, \tag{A.2.23}$$

for some $b_2, b_3 \neq 0$. Moving on, we get

$$\text{at } \mathcal{O}(\varepsilon^2): \quad \frac{\varepsilon}{2}(F_t + F F_y) \stackrel{!}{=} \varepsilon^2(b_2(c_2 F_t + a_{21} F F_y) + b_3(c_3 F_t + (a_{31} + a_{32}) F F_y)) \tag{A.2.24}$$

From this, we read off the requirements

$$c_2 = a_{21}, \quad c_3 = a_{31} + a_{32}, \quad b_2 c_2 + b_3 c_3 = \frac{1}{2}. \tag{A.2.25}$$

Note that this is not the full solution space; it does however simplify things moving forward. Concretely, it makes the right-hand side simpler; at $\mathcal{O}(\varepsilon^3)$, we find

$$\begin{aligned}
&\frac{\varepsilon^3}{6}(F_{tt} + 2F F_{ty} + F_t F_y + F F_y^2 + F^2 F_{yy}) \\
&\stackrel{!}{=} \frac{\varepsilon^3}{2} \left[b_2(c_2^2 F_{tt} + 2c_2 a_{21} F F_{ty} + a_{21}^2 F^2 F_{yy}) + b_3(2a_{32} c_2 F_t F_y + 2a_{32} a_{21} F F_y^2 \right. \\
&\quad \left. + c_3^2 F_{tt} + 2c_3(a_{31} + a_{32}) F F_{ty} + (a_{31} + a_{32})^2 F^2 F_{yy}) \right] \\
&= \frac{\varepsilon^3}{2} [b_2 c_2^2 (F_{tt} + 2F F_{ty} + F^2 F_{yy}) + b_3 (2a_{32} c_2 (F_t F_y + F F_y^2) + c_3^2 (F_{tt} + 2F F_{ty} + F^2 F_{yy}))]
\end{aligned} \tag{A.2.26}$$

This implies

$$b_2 c_2^2 + b_3 c_3^2 = \frac{1}{3} \quad \text{and} \quad 2b_3 a_{32} c_2 = \frac{1}{3}. \tag{A.2.27}$$

At this point, it is most convenient to introduce $c_2 = \alpha$ and $c_3 = \beta$ as freely choosable constants. In doing so, we get the system of equations

$$b_1 = 1 - b_2 - b_3, \quad \alpha b_2 + \beta b_3 = \frac{1}{2}, \quad \alpha^2 b_2 + \beta^2 b_3 = \frac{1}{3}, \quad a_{32} = \frac{1}{6\alpha b_3}. \tag{A.2.28}$$

This is solved by

$$\begin{aligned}
b_1 &= -\frac{3\alpha(1-2\beta)+3\beta-2}{6\alpha\beta}, & b_2 &= \frac{2-3\beta}{6\alpha(\alpha-\beta)}, \\
b_3 &= -\frac{2-3\alpha}{6\beta(\alpha-\beta)}, & a_{32} &= -\frac{\beta(\alpha-\beta)}{\alpha(2-3\alpha)}.
\end{aligned} \tag{A.2.29}$$

The full RK3 integration procedure thus reads

$$y(t_0 + \varepsilon) \approx y_0 + \varepsilon \left(-\frac{3\alpha(1-2\beta)+3\beta-2}{6\alpha\beta} k_1 + \frac{2-3\beta}{6\alpha(\alpha-\beta)} k_2 - \frac{2-3\alpha}{6\beta(\alpha-\beta)} k_3 \right) + \mathcal{O}(\varepsilon^4) \tag{A.2.30}$$

where

$$\begin{aligned}
k_1 &= F(t_0, y_0), \\
k_2 &= F(t_0 + \varepsilon\alpha, y_0 + \varepsilon\alpha k_1) \\
k_3 &= F\left(t_0 + \varepsilon\beta, y_0 + \varepsilon \left[\left(\beta + \frac{\beta(\alpha-\beta)}{\alpha(2-3\alpha)} \right) k_1 - \frac{\beta(\alpha-\beta)}{\alpha(2-3\alpha)} k_2 \right] \right).
\end{aligned} \tag{A.2.31}$$

A common choice for the quadrature points is $\alpha = \frac{1}{2}$, $\beta = 1$. This leads to

$$\begin{aligned}
b_1 &= \frac{1}{6}, & b_2 &= \frac{2}{3}, & b_3 &= \frac{1}{6}, \\
c_1 &= 0, & c_2 &= \frac{1}{2}, & c_3 &= 1 \\
a_{21} &= \frac{1}{2}, & a_{31} &= -1 & a_{32} &= 2.
\end{aligned} \tag{A.2.32}$$

When inserted, this yields the scheme

$$y(t_0 + \varepsilon) = y_0 + \varepsilon \left(\frac{1}{6} k_1 + \frac{2}{3} k_2 + \frac{1}{6} k_3 \right) + \mathcal{O}(\varepsilon^4) \tag{A.2.33}$$

with

$$\begin{aligned}
k_1 &= F(t_0, y_0), \\
k_2 &= F\left(t_0 + \frac{\varepsilon}{2}, y_0 + \frac{\varepsilon}{2} k_1\right), \\
k_3 &= F(t_0 + \varepsilon, y_0 - \varepsilon k_1 + 2\varepsilon k_2)
\end{aligned} \tag{A.2.34}$$

A.3 Courant-Friedrichs-Lewy (CFL) Conditions for PDEs

A.3.1 Example: Parabolic Heat Equation

In this section, we derive the CFL conditions for various discretisations of the parabolic heat equation,

$$\partial_t \phi = \alpha \Delta \phi \tag{A.3.1}$$

where $\alpha > 0$ is the diffusion coefficient and $\phi = \phi(t, \mathbf{x})$, $t \in \mathbb{R}$, $\mathbf{x} \in \mathbb{R}^d$ is the dynamical variable.

A.3.1.1 7-Point Laplacian Explicit Euler CFL Condition

For this first derivation, we consider the simplest discretisation of equation (A.3.1), taking the forward difference for the left-hand side, and the Laplacian stencil built from the second derivative stencil (A.1.4) with equal grid spacing Δx for all axes. In d dimensions, this amounts to

$$\frac{\phi(t + \Delta t, \mathbf{x}) - \phi(t, \mathbf{x})}{\Delta t} = \alpha \sum_{i=1}^d \frac{\phi(t, \mathbf{x} + \mathbf{e}_i \Delta x) - 2\phi(t, \mathbf{x}) + \phi(t, \mathbf{x} - \mathbf{e}_i \Delta x)}{\Delta x^2}, \quad (\text{A.3.2})$$

or when rearranged for $\phi(t + \Delta t, \mathbf{x})$,

$$\phi(t + \Delta t, \mathbf{x}) = \phi(t, \mathbf{x}) + C \sum_{i=1}^d [\phi(t, \mathbf{x} + \mathbf{e}_i \Delta x) - 2\phi(t, \mathbf{x}) + \phi(t, \mathbf{x} - \mathbf{e}_i \Delta x)]. \quad (\text{A.3.3})$$

where we defined $C := \frac{\alpha \Delta t}{\Delta x^2}$. The goal is to ensure C is such that the discrete linear operator acting on the right-hand side has no exponentially growing modes, i.e. that all its eigenvalues are non-negative. Since any discrete function $\psi(\mathbf{x}) = \phi(t, \mathbf{x})$ can be decomposed into functions of the form $e^{i\mathbf{k} \cdot \mathbf{x}}$, we make the ansatz for a mode

$$\phi(t, \mathbf{x}) = G^{t/\Delta x} e^{i\mathbf{k} \cdot \mathbf{x}} \quad (\text{A.3.4})$$

where G is the growth factor per timestep, fixed by the discretised equation (A.3.3). For the evolution to be stable, we must have $|G| \leq 1$ —let us examine how these modes evolve. Inserting into the left-hand side, we get

$$\phi(t, \mathbf{x}) = G^{(t/\Delta x)+1} e^{i\mathbf{k} \cdot \mathbf{x}} = G\phi(t, \mathbf{x}); \quad (\text{A.3.5})$$

hence the name *growth factor*. For the right-hand side, we expand

$$\begin{aligned} & \phi(t, \mathbf{x}) + C \sum_{i=1}^d [\phi(t, \mathbf{x} + \mathbf{e}_i \Delta x) - 2\phi(t, \mathbf{x}) + \phi(t, \mathbf{x} - \mathbf{e}_i \Delta x)] \\ &= \phi(t, \mathbf{x}) + CG^{t/\Delta x} \sum_{i=1}^d [e^{i\mathbf{k} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot \mathbf{e}_i \Delta x} - 2e^{i\mathbf{k} \cdot \mathbf{x}} + e^{i\mathbf{k} \cdot \mathbf{x}} e^{-i\mathbf{k} \cdot \mathbf{e}_i \Delta x}] \\ &= \phi(t, \mathbf{x}) \left(1 + C \sum_{i=1}^d \underbrace{[e^{ik_i \Delta x} - 2 + e^{-ik_i \Delta x}]}_{=-4 \sin^2(k_i \Delta x)} \right) \\ &= \phi(t, \mathbf{x}) \left(1 - 4C \sum_{i=1}^d \sin^2(k_i \Delta x) \right). \end{aligned} \quad (\text{A.3.6})$$

Equating both sides and dividing by $\phi(t, \mathbf{x}) \neq 0$, we establish

$$G = 1 - 4C \sum_{i=1}^d \sin^2(k_i \Delta x) \quad (\text{A.3.7})$$

The condition that $|G| \leq 1$ is satisfied if $G \leq 1$ and $G \geq -1$. The former is always satisfied since $\sin^2 \geq 0$, but the latter introduces constraints on C —and by that, on the relationship between Δt , Δx and α . Concretely, the right-hand side attains a global minimum where $k_i = \frac{\pi}{2\Delta x}$, $i = 1, \dots, d$. With this value for $\mathbf{k}$, we obtain

$$-1 \stackrel{!}{\leq} G = 1 - 4Cd \Leftrightarrow \frac{1}{2d} \geq C = \frac{\alpha \Delta t}{\Delta x^2}. \quad (\text{A.3.8})$$

We can turn this into a condition on the timestep Δt , which leads us to the CFL condition for this discrete stepping operator,

$$\Delta t \leq \frac{\Delta x^2}{2\alpha d}. \quad (\text{A.3.9})$$

A.3.1.2 7-Point Laplacian Implicit Euler CFL Condition

In the implicit case, one replaces the forward difference in time with the backwards difference, or equivalently, evaluates the right-hand side of the equation at $t + \Delta t$ instead of t . That is, the heat equation is discretised as

$$\frac{\phi(t + \Delta t, \mathbf{x}) - \phi(t, \mathbf{x})}{\Delta t} = \alpha \sum_{i=1}^d \frac{\phi(t + \Delta t, \mathbf{x} + \mathbf{e}_i \Delta x) - 2\phi(t + \Delta t, \mathbf{x}) + \phi(t + \Delta t, \mathbf{x} - \mathbf{e}_i \Delta x)}{\Delta x^2}. \quad (\text{A.3.10})$$

Inserting the ansatz $\phi(t, \mathbf{x}) = G^{t/\Delta t} e^{i\mathbf{k} \cdot \mathbf{x}}$ and identifying terms proceeds largely identically to the explicit case, with the difference that the Laplacian stencil terms now carry an additional factor of G as they are evaluated at $t + \Delta t$ instead of t . This has as consequence that now,

$$G = 1 - 4GC \sum_{i=1}^d \sin^2(k_i \Delta x / 2), \quad (\text{A.3.11})$$

which rearranges to

$$G = \frac{1}{1 + 4C \sum_{i=1}^d \sin^2(k_i \Delta x / 2)}. \quad (\text{A.3.12})$$

The implications of this result are profound. As long as $C \geq 0$, we have $|G| \leq 1$ —this means that the evolution is unconditionally stable; it allows for an arbitrarily large timestep Δt .

This, however, does not mean that one should opt to just do one's entire simulation in a single step. Although the discrete evolution is *stable* for any $\Delta t > 0$, the error of a step with the given choices of stencils is $\mathcal{O}(\Delta t + \Delta x^2)$. Increasing Δt hence also increases the error—just as increasing Δx in the explicit case (to allow for larger Δt) can destroy the evolution due to a lack of resolution. Because of this, the initially perceived superiority of implicit over explicit time-stepping becomes more subtle; if error requirements pose harsher constraints on Δt than the explicit CFL condition does, then explicit stepping is preferable over implicit as it is cheaper computationally.

A.3.1.3 27-point Isotropic Laplacian Explicit Euler CFL Condition

We now move on to derive the CFL condition for the explicit Euler case where on the right-hand side, the stencil (A.1.11) is employed. Concretely, we thus consider the stepping scheme

$$\phi(t + \Delta t, \mathbf{x}) = \phi(t, \mathbf{x}) + C(-164 \cdot (\text{center}) + 30 \cdot (\text{faces}) - 2 \cdot (\text{edges}) + 1 \cdot (\text{corners})), \quad (\text{A.3.13})$$

where $C = \frac{\alpha \Delta t}{26 \Delta x^2}$. Similar steps to the previous two derivations, together with the fact that the stencil applied to a constant yields 0, leads to

$$G(\mathbf{k}) = 1 - 4C \left(30 \sum_{i=1}^3 \sin^2(k_i \Delta x / 2) - 2 \sum_{i < j} [\sin^2((k_i + k_j) \Delta x / 2) + \sin^2((k_i - k_j) \Delta x / 2)] \right. \\ \left. + \sum_{\sigma_y, \sigma_z \in \{\pm 1\}} \sin^2((k_x + \sigma_y k_y + \sigma_z k_z) \Delta x / 2) \right) \quad (\text{A.3.14})$$

Denoting the large parentheses by $S(\mathbf{k})$, we obtain the stability condition

$$-1 \leq 1 - 4CS(\mathbf{k}) \Leftrightarrow S(\mathbf{k}) \leq \frac{1}{2C}. \quad (\text{A.3.15})$$

We are thus left to find the maximum of $S(\mathbf{k})$. Introducing the auxiliary variables

$$\xi_i = \frac{k_i \Delta x}{2}, \quad u_i = \sin^2(\xi_i), \quad (\text{A.3.16})$$

we can rewrite the sum over the faces as

$$\sum_{i=1}^3 \sin^2(k_i \Delta x / 2) = \sum_i u_i. \quad (\text{A.3.17})$$

For the sum over the edges, we make use of

$$\begin{aligned} \sin^2(\xi_i + \xi_j) + \sin^2(\xi_i - \xi_j) &= 2(\sin^2 \xi_i \cos^2 \xi_j + \cos^2 \xi_i \sin^2 \xi_j) \\ &= 2(u_i(1 - u_j) + (1 - u_i)u_j) \\ &= 2(u_i + u_j - 2u_i u_j) \end{aligned} \quad (\text{A.3.18})$$

so that

$$\begin{aligned} \sum_{i < j} [\sin^2((k_i + k_j) \Delta x / 2) + \sin^2((k_i - k_j) \Delta x / 2)] &= 2 \sum_{i < j} [u_i + u_j - 2u_i u_j] \\ &= 4 \sum_i u_i - 4 \sum_{i < j} u_i u_j. \end{aligned} \quad (\text{A.3.19})$$

For the corner terms, we consider

$$\begin{aligned} \sin(\xi_x + \sigma_y \xi_y + \sigma_z \xi_z) &= \sin(\xi_x + \sigma_y \xi_y) \cos(\xi_z) + \sigma_z \cos(\xi_x + \sigma_y \xi_y) \sin(\xi_z) \\ &= \sin(\xi_x) \cos(\xi_y) \cos(\xi_z) + \sigma_y \cos(\xi_x) \sin(\xi_y) \cos(\xi_z) \\ &\quad + \sigma_z \cos(\xi_x) \cos(\xi_y) \sin(\xi_z) - \sigma_y \sigma_z \sin(\xi_x) \sin(\xi_y) \sin(\xi_z) \end{aligned} \quad (\text{A.3.20})$$

After squaring, this turns into a horrible mess, but since we are summing over all values of $\sigma_y, \sigma_z \in \{\pm 1\}$, any term linear in one of the σ will cancel in the sum. That is, only the squares of the individual summands appear in the final result, so that

$$\begin{aligned} \sum_{\sigma_y, \sigma_z \in \{\pm 1\}} \sin^2((k_x + \sigma_y k_y + \sigma_z k_z) \Delta x / 2) &= 4(\sin^2 \xi_x \cos^2 \xi_y \cos^2 \xi_z + \cos^2 \xi_x \sin^2 \xi_y \cos^2 \xi_z \\ &\quad + \cos^2 \xi_x \cos^2 \xi_y \sin^2 \xi_z - \sin^2 \xi_x \sin^2 \xi_y \sin^2 \xi_z) \\ &= 4(u_x(1 - u_y)(1 - u_z) + (1 - u_x)u_y(1 - u_z) \\ &\quad + (1 - u_x)(1 - u_y)u_z + u_x u_y u_z) \\ &= 4 \left(\sum_i u_i - 2 \sum_{i < j} u_i u_j + 4u_x u_y u_z \right) \end{aligned} \quad (\text{A.3.21})$$

Hence, $S(\mathbf{k})$, now as a function of $\mathbf{u} = (u_x, u_y, u_z)$ reads

$$\begin{aligned} S(\mathbf{u}) &= 30 \sum_i u_i - 2 \left(4 \sum_i u_i - 4 \sum_{i < j} u_i u_j \right) + 4 \left(\sum_i u_i - 2 \sum_{i < j} u_i u_j + 4u_x u_y u_z \right) \\ &= 26(u_x + u_y + u_z) + 16u_x u_y u_z. \end{aligned} \quad (\text{A.3.22})$$

Since $u_i \in [0, 1]$, S clearly takes its maximum where $u_x = u_y = u_z = 1$, yielding

$$\max_{[0,1]^3} S(\mathbf{u}) = 94. \quad (\text{A.3.23})$$

Thus, the condition (A.3.15) is turned into

$$94 \leq \frac{1}{2C} \Leftrightarrow \frac{\alpha \Delta t}{26 \Delta x^2} = C \leq \frac{1}{188} \quad (\text{A.3.24})$$

Rearranging for Δt yields the final condition

$$\Delta t \leq \frac{13}{94} \frac{\Delta x^2}{\alpha}. \quad (\text{A.3.25})$$

Note that

$$\frac{13}{94} \approx 0.13892..., \quad (\text{A.3.26})$$

making the 27-point stencil CFL condition slightly more restrictive than for the 7-point case, where the factor in front of $\Delta x^2/\alpha$ is

$$\frac{1}{2d} = \frac{1}{6} \approx 0.166\bar{6}. \quad (\text{A.3.27})$$

This trade-off is typically worth it, given that the isotropy of the error is much higher.

A.3.1.4 27-point Isotropic Laplacian Implicit Euler CFL Condition

In the implicit case of the preceding section, where the right-hand side of equation (A.3.13) is evaluated at $t + \Delta t$ instead, we obtain

$$G = 1 - 4GCS(\mathbf{k}). \quad (\text{A.3.28})$$

Solving for G , we find

$$G = \frac{1}{1 + 4CS(\mathbf{k})}. \quad (\text{A.3.29})$$

As we have shown before,

$$S(\mathbf{k}(\mathbf{u})) = 26(u_x + u_y + u_z) + 16u_x u_y u_z \in [0, 94], \quad (\text{A.3.30})$$

so that $|G| \leq 1$ is always satisfied and hence, the implicit Euler procedure remains unconditionally stable even with the isotropic stencil for the Laplacian.

A.3.2 Example: Hyperbolic Wave Equation

The derivation of CFL conditions for the wave equation,

$$\square \phi = 0 \Leftrightarrow \partial_t^2 \phi = c^2 \Delta \phi, \quad (\text{A.3.31})$$

where c is the propagation velocity. The simplest discretisation for this equation (and the only we will consider here) is

$$\frac{\phi(t + \Delta t, \mathbf{x}) - 2\phi(t, \mathbf{x}) + \phi(t - \Delta t, \mathbf{x})}{\Delta t^2} = c^2 \sum_i \frac{\phi(t, \mathbf{x} + \mathbf{e}_i \Delta x) - 2\phi(t, \mathbf{x}) + \phi(t, \mathbf{x} - \mathbf{e}_i \Delta x)}{\Delta x^2} \quad (\text{A.3.32})$$

This form, as is, is unsuited for a time-stepping procedure, but we can derive stability conditions from it. Again employing the ansatz $\phi(t, \mathbf{x}) = G^{t/\Delta t} e^{i\mathbf{k} \cdot \mathbf{x}}$ for an amplification number G , we can rewrite the above as

$$\begin{aligned}
(G - 2 + G^{-1})\phi(t, \mathbf{x}) &= C^2 \sum_i (e^{ik_i \Delta x} - 2 + e^{-ik_i \Delta x})\phi(t, \mathbf{x}) \\
&= -4C^2 \sum_i \sin^2(k_i \Delta x/2)\phi(t, \mathbf{x}) \\
&=: -C^2 S(\mathbf{k})\phi(t, \mathbf{x}).
\end{aligned} \tag{A.3.33}$$

Rearranging and multiplying by G leads to

$$G^2 - 2\left(1 - \frac{1}{2}C^2 S(\mathbf{k})\right)G + 1 = 0 \tag{A.3.34}$$

The allowed amplification factors are the two roots $G_{\pm}$ of this polynomial in G . Since the constant term is 1, they must satisfy $G_+ G_- = 1$, whence we must have $|G_{\pm}| = 1$ to satisfy the no-growth condition $|G| \leq 1$.

For a general polynomial of this form,

$$G^2 - 2\beta G + 1 = 0, \quad \beta \in \mathbb{R}, \tag{A.3.35}$$

the roots read

$$G_{\pm} = \beta \pm \sqrt{\beta^2 - 1}. \tag{A.3.36}$$

If $\beta^2 > 1$, we have strictly real and distinct roots, of which at least one has a magnitude greater than 1. Thus, we must have $\beta^2 \leq 1$, or equivalently, $-1 \leq \beta \leq 1$. Inserting the β from equation (A.3.34), this turns into

$$-1 \leq 1 - \frac{1}{2}C^2 S(\mathbf{k}) \leq 1. \tag{A.3.37}$$

The right-hand bound holds trivially; the left-hand bound requires

$$C^2 S(\mathbf{k}) \leq 4 \quad \Rightarrow \quad C^2 S_{\max} \leq 4 \tag{A.3.38}$$

Since $S_{\max} = 4d$, this yields the final CFL condition,

$$\frac{c^2 \Delta t^2}{\Delta x^2} = C^2 \leq \frac{1}{d} \quad \Leftrightarrow \quad \Delta t \leq \frac{\Delta x}{c\sqrt{d}}. \tag{A.3.39}$$

This has a very nice physical interpretation: the distance travelled by a wave within one timestep, $c\Delta t$, must not exceed a value proportional to the grid spacing, Δx . Since the proportionality factor of $1/\sqrt{d}$ is less than 1, this means that the numerical domain of dependence is contained in the physical domain of dependence.

A.4 Implicit ODE and PDE Solvers

In this section, we consider implicit numerical integration schemes for differential equations of the form

$$\partial_t \phi = F(t, \phi), \tag{A.4.1}$$

where F is either a function or a functional of ϕ , valued in the same space as ϕ itself. That is, we consider either ODEs or PDEs, for which we develop implicit integration schemes. Although these terms have come up in the above, we should first clarify more precisely what is meant by an *implicit* integration scheme. We do this by specifying what an *explicit* integration scheme is; an implicit scheme is anything that is not explicit.

Formally, we can integrate equation (A.4.1) from some initial time t to some later time $t + \Delta t$ as

$$\phi(t + \Delta t) = \phi(t) + \int_t^{t+\Delta t} F(t', \phi(t')) dt'. \quad (\text{A.4.2})$$

Of course, this is not some magic solution recipe—we have simply turned a differential equation into an integro-differential equation. However now, an integral over t is involved, and we can turn to quadratures to approximate it.

Let us assume that we have a functional $G(t, \phi)$ (typically a quadrature rule depending on evaluations of ϕ at discrete points in time) which approximates this integral up to some order k in Δt , that is

$$\int_t^{t+\Delta t} F(t', \phi(t')) dt' = G(t, \phi) + \mathcal{O}(\Delta t^k), \quad (\text{A.4.3})$$

or equivalently for the time-step,

$$\phi(t + \Delta t) = \phi(t) + G(t, \phi) + \mathcal{O}(\Delta t^k). \quad (\text{A.4.4})$$

Given this setup, we call the stepping scheme *explicit* if $G(t, \phi)$ depends only on values of ϕ known at time t —i.e. only on values lying in the past of t . If G involves e.g. $\phi(t + \Delta t)$, it is called *implicit*.

An important consequence of G depending on field configurations other than $\phi(t)$ is that the stepping implemented in code,

$$\phi(t + \Delta t) \approx \phi(t) + G(t, \phi), \quad (\text{A.4.5})$$

must now be *solved for* $\phi(t + \Delta t)$, rather than being a direct assignment operation.

A.5 Implicit ODE and PDE Solvers

In this section, we consider numerical integration schemes for differential equations of the form

$$\partial_t \phi = F(t, \phi), \quad (\text{A.5.1})$$

where F is either a function (for ODEs) or an operator/functional (for PDEs) acting on ϕ .

Formally, integrating equation (A.5.1) over a time interval $[t, t + \Delta t]$ yields the exact integral equation

$$\phi(t + \Delta t) = \phi(t) + \int_t^{t+\Delta t} F(t', \phi(t')) dt'. \quad (\text{A.5.2})$$

Numerical time-stepping algorithms replace this integral with a discrete quadrature rule. Let $\phi^n \approx \phi(n\Delta t)$ denote the discrete numerical solution at step n . A general single-step time-integration scheme approximates the integral via a function G ,

$$\phi^{n+1} \approx \phi^n + \Delta t G(t_n, \Delta t, \phi^n, \phi^{n+1}), \quad (\text{A.5.3})$$

up to some local truncation error $\mathcal{O}(\Delta t^{k+1})$, yielding a scheme of order k .

We classify the integration scheme based on how G depends on the state vector:

- (i) *Explicit*: G depends solely on known state values from the current or previous time steps $(\phi^n, \phi^{n-1}, \dots)$. The update step for ϕ^{n+1} is a direct assignment operation.

- (ii) *Implicit*: G depends on the unknown state ϕ^{n+1} at the new time level (or intermediate stage evaluations requiring ϕ^{n+1}).

Consequently, for an implicit scheme, the update formula cannot be evaluated directly; instead, ϕ^{n+1} must be obtained by solving a system of algebraic equations (linear or non-linear) at every time step.

As the name would suggest, explicit/forward Euler,

$$\phi(t + \Delta t) = \phi(t) + \Delta t F(t, \phi(t)) + \mathcal{O}(\Delta t) \quad (\text{A.5.4})$$

is an example of an explicit integration scheme. Further examples are the Runge-Kutta integrators we considered in Section A.2. Another class of examples, which incorporates past values $\phi(t - \Delta t), \phi(t - 2\Delta t), \dots$ exist, and are known as *Adams-Bashforth schemes*.

A.5.1 Example: Backwards Euler

In this section, we introduce the most basic of implicit time integration schemes, the *backwards/implicit Euler method*. Although it has a rather large error, it will serve well in introducing a number of concepts, and provide a reference to compare more elaborate methods against.

The arguably simplest way to approximate the integral

$$\int_t^{t+\Delta t} F(t', \phi(t')) dt' \quad (\text{A.5.5})$$

is to take the value of ϕ at one of the boundary times, $t, t + \Delta t$, and multiply it by the width Δt of the integration interval. Since we are interested in an implicit scheme, we take the boundary value at $t + \Delta t$, that is,

$$\int_t^{t+\Delta t} F(t', \phi(t')) dt' = \Delta t F(t + \Delta t, \phi(t + \Delta t)) + \mathcal{O}(\Delta t^2) \quad (\text{A.5.6})$$

Neglecting to explicitly write the error term, this leads to the time-stepping scheme

$$\phi(t + \Delta t) = \phi(t) + \Delta t F(t + \Delta t, \phi(t + \Delta t)). \quad (\text{A.5.7})$$

Clearly, this is an equation to be solved for $\phi(t + \Delta t)$. In the case of F being a function, it is purely algebraic; if F however depends on derivatives of ϕ , it is a differential equation.

Let us go over some examples below.

- (i) **Exponential Decay**: For $\phi : \mathbb{R} \rightarrow \mathbb{R}$, we consider the ODE

$$\partial_t \phi(t) = -\phi(t). \quad (\text{A.5.8})$$

Clearly, the solution space is given by $\phi(t) = \phi_0 e^{-t}$, $\phi_0 \in \mathbb{R}$. The implicit time-stepping scheme prescribes the update equation

$$\phi(t + \Delta t) = \phi(t) + \Delta t (-\phi(t + \Delta t)). \quad (\text{A.5.9})$$

In this case—and many other ODE cases—it is possible to explicitly solve this for $\phi(t + \Delta t)$:

$$\phi(t + \Delta t) = \frac{\phi(t)}{1 + \Delta t}. \quad (\text{A.5.10})$$

This is a recursion, which can be made closed-form as

$$\phi(n\Delta t) = \frac{\phi_0}{(1 + \Delta t)^n} \quad (\text{A.5.11})$$

As expected from the analytical solution $\phi(t) = \phi_0 e^{-t}$, this is an exponential decay; however, each step truncates the exponential series to linear order. One timestep of the analytical solution scales it by a factor

$$e^{-\Delta t} = \frac{1}{e^{\Delta t}} = \frac{1}{1 + \Delta t + \frac{1}{2}\Delta t^2 + \dots}. \quad (\text{A.5.12})$$

The implicit Euler scheme hence neglects the $\mathcal{O}(\Delta t^2)$ terms in the denominator.

Lastly, we note that the stepping scheme is unconditionally stable—for any positive Δt , equation (A.5.11) defines a stable exponential decay. The explicit/forward Euler scheme, which prescribes

$$\phi(t + \Delta t) = \phi(t) + \Delta t(-\phi(t)) = (1 - \Delta t)\phi(t), \quad (\text{A.5.13})$$

has the closed-form solution

$$\phi(n\Delta t) = (1 - \Delta t)^n \phi_0, \quad (\text{A.5.14})$$

which for $\Delta t > 1$ becomes oscillatory, and for $\Delta t > 2$ starts growing exponentially in magnitude. Further, it is the truncation of the implicit solution:

$$\frac{\phi_0}{(1 + \Delta t)^n} = (1 - \Delta t + \Delta t^2 + \dots)^n \phi_0 \approx (1 - \Delta t)^n \phi_0 \quad (\text{A.5.15})$$

What we can take away from this example is the following: implicit Euler handles exponential decay much more gracefully than explicit Euler.

(ii) **1d Heat Equation, Discretised:** We now proceed to the one-dimensional heat equation,

$$\partial_t \phi = \partial_x^2 \phi. \quad (\text{A.5.16})$$

Applying the implicit Euler method to this PDE leads to the time-step

$$\phi(t + \Delta t, x) = \phi(t, x) + \Delta t(\partial_x^2 \phi)(t + \Delta t, x). \quad (\text{A.5.17})$$

Here, separating quantities evaluated at $t + \Delta t$ to the left-hand side, we obtain the equation

$$(\phi - \Delta t \partial_x^2 \phi)(t + \Delta t, x) = \phi(t, x). \quad (\text{A.5.18})$$

This is a second-order linear ODE with constant coefficients, where the previous field state $\phi(t, x)$ acts as a source. Although technically, depending on the initial field configuration $\phi_0(x)$, we can sometimes integrate this equation for each step, let us entertain a different (and more versatile) approach.

Concretely, we discretise ϕ not just in time but also in space, with a grid spacing of Δx . Since otherwise, notation will get heavy, let us introduce the shorthand

$$\phi_i^n = \phi(n\Delta t, i\Delta x). \quad (\text{A.5.19})$$

Employing the stencil (A.1.4) for the second derivative, the discretised step equation reads

$$\phi_i^{n+1} - C(\phi_{i-1}^{n+1} - 2\phi_i^{n+1} + \phi_{i+1}^{n+1}) = \phi_i^n \quad (\text{A.5.20})$$

where $C = \Delta t / \Delta x^2$. This is a linear system of equations—let us write it in matrix-vector form. Denoting $\boldsymbol{\phi}^n = (\phi_i^n)_{i=1}^N$ for some grid size N , and additionally imposing Dirichlet boundary conditions, $\phi_{-1} = \phi_{N+1} = 0$, we can turn the step equation into

$$\mathbf{M}\boldsymbol{\phi}^{n+1} = \boldsymbol{\phi}^n \quad (\text{A.5.21})$$

with

$$\mathbf{M} = \begin{pmatrix} 1-2C & -C & & & \\ -C & 1-2C & -C & & \\ & -C & 1-2C & \ddots & \\ & & \ddots & \ddots & -C \\ & & & -C & 1-2C \end{pmatrix} \quad (\text{A.5.22})$$

The first instinct might be to make the step equation explicit by writing

$$\phi^{n+1} = \mathbf{M}^{-1} \phi^n, \quad (\text{A.5.23})$$

and precomputing the inverse $\mathbf{M}^{-1}$ once at simulation start. Although this works, this has a significant computational drawback: the matrix $\mathbf{M}$ is sparse, but $\mathbf{M}^{-1}$ is not. This makes the multiplication between $\mathbf{M}^{-1}$ and the state vector ϕ^n computationally expensive, especially for larger grids. Because of this, one usually opts to solve the linear system using other methods, such as an LDU decomposition or Gauss-Seidel.

(iii) **Heat Equation, Spectral:** We now approach the heat equation in d dimensions,

$$\partial_t \phi = \Delta \phi, \quad \Delta = \sum_{i=1}^d \partial_i^2, \quad (\text{A.5.24})$$

on the domain $\Omega = [0, \pi]^d$ with Dirichlet boundary conditions, $\phi|_{\partial\Omega} = 0$. Its implicit Euler step equation is clearly

$$\phi(t + \Delta t, \mathbf{x}) = \phi(t, \mathbf{x}) + \Delta t (\Delta \phi)(t + \Delta t, \mathbf{x}) \quad (\text{A.5.25})$$

This rearranges into an elliptic problem at each step,

$$(\phi - \Delta t \Delta \phi)(t + \Delta t, \mathbf{x}) = \phi(t, \mathbf{x}). \quad (\text{A.5.26})$$

Just as in the $d = 1$ case above, we could now discretise the Laplacian operator and proceed solving the linear system using Gauss-Seidel or any other linear solver. Writing this down explicitly is more tedious, since the index gymnastics are far more involved, and there is not much gained conceptually from doing so.

Instead, we entertain another approach of approximating the solution ϕ and by that the action of Δ on it—by means of a spectral expansion, in the special case where $d = 3$. Concretely, we define the functions

$$\psi_{ijk}(\mathbf{x}) = \sin(ix) \sin(jy) \sin(kz), \quad i, j, k \in \mathbb{N}^3. \quad (\text{A.5.27})$$

Clearly, these functions form an orthogonal basis with respect to the standard L^2 -inner product on the space of L^2 -functions on Ω satisfying the boundary conditions. We can hence express $\phi(t, \mathbf{x})$ as a sum with time-variable coefficients $c_{ijk}(t)$,

$$\phi(t, \mathbf{x}) = \sum_{i,j,k \in \mathbb{N}} c_{ijk}(t) \psi_{ijk}(\mathbf{x}). \quad (\text{A.5.28})$$

Since computers cannot deal with infinities, we must truncate the sum at some highest-frequency mode (N, N, N) , turning it into

$$\phi(t, \mathbf{x}) = \sum_{i,j,k=1}^N c_{ijk}(t) \psi_{ijk}(\mathbf{x}). \quad (\text{A.5.29})$$

One might argue that this loses information just as discretising ϕ does. However, the kind of information that is lost is different; While discretising the ϕ leaves the function values at the grid points exact but makes derivatives introduce errors, truncating the basis expansion at some finite point (typically) allows derivatives to remain exact at the cost of introducing some error in raw function values. Let us examine this. The action of the Laplacian on a basis function ψ_{ijk} is

$$\Delta\psi_{ijk} = -(i^2 + j^2 + k^2)\psi_{ijk}. \quad (\text{A.5.30})$$

Since the right-hand side involves no modes with larger indices, it can still be resolved fully by a series truncated at the mode (N, N, N) . Thus, the Laplacian applied to the truncated basis expansion of $\phi(t, \mathbf{x})$ is exact, reading

$$\Delta\phi = - \sum_{i,j,k=1}^N (i^2 + j^2 + k^2) c_{ijk} \psi_{ijk}. \quad (\text{A.5.31})$$

We can insert this expansion into equation (A.5.26) to obtain

$$\sum_{i,j,k=1}^N (c_{ijk}^{n+1} + \Delta t(i^2 + j^2 + k^2)c_{ijk}^{n+1})\psi_{ijk} = \sum_{i,j,k=1}^N c_{ijk}^n \psi_{ijk}. \quad (\text{A.5.32})$$

Since basis expansions are unique, identification of coefficients implies the update equation

$$c_{ijk}^{n+1} = \frac{c_{ijk}^n}{1 + \Delta t(i^2 + j^2 + k^2)}. \quad (\text{A.5.33})$$

This results in a geometric decay, with higher-frequency modes being suppressed more rapidly—just as expected for the heat equation.

Remark The reason we can invert equation (A.5.32) nicely for the explicit iteration step above is that the operator acting on the basis coefficient vector on the left-hand side is diagonal. In other words, the chosen basis diagonalises the Laplacian. Unfortunately, we cannot always do this; given a differential operator L , it is not always possible to find an explicit basis that diagonalises it so that the associated system

$$\partial_t \phi = L\phi \quad (\text{A.5.34})$$

can be time-integrated spectrally as cleanly as above. However, there are certain valuable properties of a basis $\{\psi_n\}$ (with n some generalised index) that typically can be achieved, which we now outline.

- The arguably most important advantage that spectral methods have over finite differences is that derivatives can be exact. For this to be the case, however, the truncation subspace $V_N = \text{span}\{\psi_1, \dots, \psi_N\}$ must be closed under differentiation. Just as we can adjust the resolution of a discretisation grid, we want to be able to adjust N freely (up to stability and error considerations), which makes closure under differentiation require

$$\partial\psi_n \in \text{span}\{\psi_1, \dots, \psi_n\} \quad \forall n \in \mathbb{N}. \quad (\text{A.5.35})$$

Here, ∂ represents any of the relevant first derivatives.

- Further, we should choose our basis such that L becomes lower triangular with respect to it on V_N . This makes solving the equivalent of equation (A.5.26) more straightforward to solve by employing a single Gauss elimination. Lower triangularity however is not a strict necessity, there are other forms where similarly efficient linear solvers exist

Polynomial bases such as Legendre, Laguerre or Chebyshev polynomials work well for this application, since differentiation reduces their degree and hence ensures closure. For periodic domains, plane waves ($e^{i\mathbf{k}\cdot\mathbf{x}}$) are highly versatile, and for infinite domains, one can either map to a finite one with a conformal transformation and use a polynomial basis or employ Hermite functions if features are highly localised around the origin.

For linear equations with constant coefficients, the exact spatial differentiation of spectral methods is a clear gain over discrete finite differences. The only errors emerge from the approximation of initial conditions when projecting onto the basis and truncating, as well as the error introduced by the time integrator. However, as soon as terms which are non-linear in the basis functions emerge, we start to introduce additional truncation error. To illustrate this, consider terms of the form

$$f\phi \quad \text{or} \quad \phi^2. \quad (\text{A.5.36})$$

When inserting basis expansions for the field ϕ and auxiliary functions such as f , products $\psi_n\psi_m$ of basis functions emerge. Although these can be expressed as a linear combination of ψ_n 's again, this may introduce contributions from modes that are above the truncation cutoff N . Concretely, consider for example the real Fourier basis,

$$\psi_n(x) = \cos(nx), \quad \chi_n(x) = \sin(nx). \quad (\text{A.5.37})$$

The product of two such basis functions yields linear combinations like

$$\begin{aligned} \psi_n(x)\psi_m(x) &= \cos(nx)\cos(mx) = \frac{1}{2}\cos((n-m)x) + \frac{1}{2}\cos((n+m)x) \\ &= \frac{1}{2}\psi_{n-m}(x) + \frac{1}{2}\psi_{n+m}(x), \\ \chi_n(x)\psi_m(x) &= \sin(nx)\cos(mx) = \frac{1}{2}\sin((n-m)x) + \frac{1}{2}\sin((n+m)x) \\ &= \frac{1}{2}\chi_{n-m}(x) + \frac{1}{2}\chi_{n+m}(x), \\ \chi_n(x)\chi_m(x) &= \sin(nx)\sin(mx) = \frac{1}{2}\cos((n-m)x) - \frac{1}{2}\cos((n+m)x) \\ &= \frac{1}{2}\psi_{n-m}(x) - \frac{1}{2}\psi_{n+m}(x). \end{aligned} \quad (\text{A.5.38})$$

This means that the product of two modes will alias as a low-frequency mode ψ_{n-m} or χ_{n-m} *as well as* a high-frequency mode ψ_{n+m} or χ_{n+m} . If $n+m > N$, the high-frequency alias is truncated, introducing an error. For this error not to be devastating, N has to be chosen large enough so that modes close to it are low in amplitude.

A.5.2 Example: Crank-Nicolson

A better way of approximating an integral than by taking one of its endpoint values multiplied by the interval width is to approximate the integrand as the linear polynomial passing through both endpoints. This is the so-called *trapezoidal* integration rule,

$$\int_a^b f(t) dt = \frac{\Delta t}{2}(f(b) + f(a)) + \mathcal{O}(\Delta t^3), \quad (\text{A.5.39})$$

which, in particular, improves the error from $\mathcal{O}(\Delta t^2)$ to $\mathcal{O}(\Delta t^3)$.

A.5.3 Example: Implicit Runge-Kutta

A.5.4 Example: IMEX Schemes (Implicit-Explicit)

A.5.5 Example: Alternating Direction Implicit (ADI)

A.6 Adaptive Mesh Refinement

A.6.1 Refinement Conditions

The goal of adaptive mesh refinement (AMR) is to increase the resolution of a simulation wherever there are features in the field configuration which cannot be resolved adequately at the current resolution. Hence, we need a predicate to decide whether a grid cell should be refined or not; for this, we need to be able to detect features.

A first intuition for how to detect features might be something gradient-based—something like

$$\text{refine if } |\nabla\phi| > C \quad (\text{A.6.1})$$

for some threshold value C . Unfortunately, this has multiple issues:

- (i) It is entirely decoupled from the current grid resolution. If large enough, the same gradient value will always lead to refinement, regardless of what resolution we already have.
- (ii) Further, the gradient is scale-dependent. A rescaling $\phi \rightarrow \lambda\phi$ for some real constant $\lambda > 0$ does not affect the accuracy of a finite-difference stencil, since it simply gets scaled by the same factor. However, since $|\nabla\phi| \rightarrow \lambda|\nabla\phi|$, the condition depends upon the scale of ϕ .
- (iii) For affine linear functions $\phi(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x} + \mathbf{b}$, finite difference approximations of its derivatives are exact. However, since $\nabla\phi = \mathbf{a}$, so we refine if $|\mathbf{a}| > C$. This is bad: we do not need to increase the resolution for field configurations where the error of finite difference approximations vanishes already.

Let us address these issues in order. The first is rather simple to fix: we simply multiply by the grid spacing Δx , and turn the refinement condition into

$$\Delta x |\nabla\phi| > C. \quad (\text{A.6.2})$$

Effectively, this removes the division by the grid spacing in the finite difference approximation. Concretely, in the one-dimensional case, we have

$$C < \Delta x |\nabla\phi| \approx \Delta x \frac{\phi(x + \Delta x) - \phi(x - \Delta x)}{2\Delta x} = \frac{1}{2}(\phi(x + \Delta x) - \phi(x - \Delta x)). \quad (\text{A.6.3})$$

We hence respect the grid spacing now; we refine if the difference between neighbouring cells' values becomes too large. This is already a much more sensible condition—we no longer refine indefinitely, but rather, until the change from cell to cell is small enough.

To address the second issue, (ii), we might think to rescale the gradient by $|\phi|$, turning it into

$$\frac{\Delta x |\nabla\phi|}{|\phi|} > C. \quad (\text{A.6.4})$$

Although this is now invariant under $\phi \rightarrow \lambda\phi$, we have introduced two new issues: a potential division by zero, and a sensitivity to shift, $\phi \rightarrow \phi + \alpha$. The division by zero can be remedied by adding a typical scale/noise floor ϕ_0 of the field ϕ to the denominator, turning it into

$$\frac{\Delta x |\nabla\phi|}{|\phi| + \phi_0} > C. \quad (\text{A.6.5})$$

This is resolution- and scale-invariant, so we have solved (i) and (ii), but (iii) is still unresolved—it got even worse—and we are now sensitive to shifts, $\phi \rightarrow \phi + \alpha$.

So, let us try to address (iii), together with this new issue. The shift-covariance issue is simple enough to deal with; any derivative of ϕ is shift-invariant, so we only use its derivatives in our condition. We are left to address (iii). Since we would like our condition's left-hand side

to ideally evaluate to zero for affine-linear functions, we are bound to make it proportional to *second* derivatives of ϕ . The likely most infamous scalar second derivative of a function is its Laplacian, $\Delta\phi$. Being a second derivative, it maps affine-linear functions $\mathbf{a} \cdot \mathbf{x} + \mathbf{b}$ to zero, but unfortunately, it also annihilates the entire class of harmonic functions as well. For example, the function $\phi(x, y) = e^x \sin y$ is harmonic, but its finite difference approximations of derivatives are not trivially exact. Thus, using $\Delta\phi$ is not viable.

The most general second derivative of ϕ is its Hessian, which we denote by

$$H = \left[\frac{\partial^2 \phi}{\partial x^i \partial x^j} \right]_{i,j=1}^d. \quad (\text{A.6.6})$$

We would like to build a scalar from it that is sensitive to any second-order features, and which is isotropic. Under an orthogonal transformation $\mathbf{x} \rightarrow R\mathbf{x}$, with $R^\top R = I$, the Hessian transforms as $H \rightarrow RHR^\top$. Unfortunately, we have already ruled out $\text{tr} H = \Delta\phi$ as an option, so the next-best isotropic scalar is the Frobenius norm

$$\|H\|_F^2 = \text{tr}(H^\top H). \quad (\text{A.6.7})$$

It is indeed isotropic, since

$$\text{tr}(H^\top H) \rightarrow \text{tr}\left((RHR^\top)^\top RHR^\top\right) = \text{tr}(RH^\top R^\top RHR^\top) = \text{tr}(H^\top H). \quad (\text{A.6.8})$$

In e.g. $d = 3$, we can write it out explicitly as

$$\|H\|_F^2 = \phi_{xx}^2 + \phi_{yy}^2 + \phi_{zz}^2 + 2\phi_{xy}^2 + 2\phi_{yz}^2 + 2\phi_{zx}^2. \quad (\text{A.6.9})$$

Since all terms are non-negative, any non-zero second-order derivative will be picked up by $\|H\|_F$. Moreover, due to isotropy, it is agnostic to the feature's orientation—we have thus found an ingredient for a refinement condition solving (iii).

Taking into account everything we have established in the above, a good refinement condition is

$$\text{refine if } \frac{\Delta x^4 \|H\|_F^2}{\Delta x^2 |\nabla \phi|^2 + \phi_0^2} > C. \quad (\text{A.6.10})$$

This is a generalisation of the Löhner error estimate. At this point it is worth considering its behaviour in different regimes. In a region of small gradients—where $\Delta x^2 |\nabla \phi|^2 \ll \phi_0^2$, such as at the top of a bell curve—the condition collapses to

$$\frac{\Delta x^4}{\phi_0^2} \|H\|_F^2 > C. \quad (\text{A.6.11})$$

This means that refinement is proportional to the curvature $\|H\|_F^2$. In the other limiting case— $\Delta x^2 |\nabla \phi|^2 \gg \phi_0^2$, such as near shocks or wavefronts—the condition turns into

$$\Delta x^2 \frac{\|H\|_F^2}{|\nabla \phi|^2} > C. \quad (\text{A.6.12})$$

This is a measure for the relative rate of change of the gradient, loosely to be interpreted as $\nabla \log |\nabla \phi|$.

A.6.2 Subcycling

B Numerical Relativity

B.1 Conventions

In the following, the spacetime metric g always has signature $(-+++)$. The induced metric γ and conformal metric $\tilde{\gamma}$ are Riemannian, that is, of signature $(+++)$. The Riemann tensor has the sign convention

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z. \quad (\text{B.1.1})$$

We use the Einstein equations in the form

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}, \quad (\text{B.1.2})$$

that is, with $c = G = 1$.

B.2 The 3 + 1-Formalism

B.2.1 Foliations and Projectors

On a Lorentzian 3-manifold $(\mathcal{M}, g)$, given a function $t : \mathcal{M} \rightarrow \mathbb{R}$ such that $g(dt, dt) \leq 0$ everywhere, we call the covering of $\mathcal{M}$ by the sets

$$\Sigma_{t_0} = \{p \in \mathcal{M} \mid t(p) = t_0\} \quad (\text{B.2.1})$$

a *spacelike foliation* of $(\mathcal{M}, g)$. If $(\mathcal{M}, g)$ is time-orientable and dt future-oriented, then the foliation $\{\Sigma_t\}$ defines a unique timelike unit normal vector field n , perpendicular to the foliation everywhere, by

$$n = -\alpha dt^\sharp, \quad (\text{B.2.2})$$

where $\alpha > 0$ is a function fixed by the normalisation condition

$$-1 = g(n, n) = \alpha^2 g^{tt} \Leftrightarrow g^{tt} = -\frac{1}{\alpha^2}. \quad (\text{B.2.3})$$

In components, we have

$$n_\mu = -\alpha \delta_\mu^t, \quad n^\mu = -\alpha g^{\mu t}. \quad (\text{B.2.4})$$

The vector n can be used to define an induced metric γ / a projection operator P ,

$$\gamma = g + n^\flat \otimes n^\flat \Leftrightarrow P = \delta + n \otimes n^\flat \quad (\text{B.2.5})$$

or equivalently in components,

$$\gamma_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu \Leftrightarrow P^\mu{}_\nu = \delta^\mu{}_\nu + n^\mu n_\nu. \quad (\text{B.2.6})$$

Note: although related by metric-induced isomorphism, the fully contra- and covariant object is denoted by $\gamma_{\mu\nu}$ and $\gamma^{\mu\nu}$, respectively, whereas the (1,1)-tensor is denoted P . This is because in its fully contra- and covariant forms, it is best interpreted as an induced metric, whereas the (1,1)-form is better interpreted as a projector.

This projector has the following properties:

- (i) $P^\mu{}_\nu n^\nu = 0$;
- (ii) If $g(X, n) = 0$ ($\Leftrightarrow X \in \Gamma(T\Sigma)$) then $P^\mu{}_\nu X^\nu = X^\mu$;
- (iii) $P^\mu{}_\nu P^\nu{}_\lambda = P^\mu{}_\lambda$;

(iv) $\gamma_{\mu\nu} = \gamma_{\nu\mu}$;

(v) $P^\mu{}_\mu = 3$.

(vi) For $X, Y \in \Gamma(T\Sigma)$, it holds that $g(X, Y) = \gamma(X, Y)$.

These properties establish that pointwise, $P : T_p \mathcal{M} \rightarrow T_p \Sigma_{t(p)}$ is a surjective orthogonal projection. In particular, $P|_{T\Sigma} = \text{id}_{T\Sigma}$, and $P|_{N\Sigma} = 0$. The last property in the above shows that γ is indeed the induced metric on Σ when restricting arguments to $T\Sigma$.

At this point, we introduce some additional notation. Given a tensor $T \in T_p^{(r,s)} \mathcal{M}$, its projection $PT \in T_p^{(r,s)} \Sigma$ is defined by

$$(PT)^{\mu\dots}_{\nu\dots} = P^\mu{}_\lambda \dots P^\rho{}_\nu \dots T^{\lambda\dots}_{\rho\dots}, \quad (\text{B.2.7})$$

that is, PT has as components those of T contracted with a projector on each index. In this notation we have, for example,

$$Pn = 0 \quad \text{and} \quad Pg = \gamma. \quad (\text{B.2.8})$$

B.2.2 Extrinsic Curvature

In the codimension 1 case, the extrinsic curvature tensor is defined by

$$K(X, Y) = g(n, \nabla_{PX} PY) \Leftrightarrow K_{\mu\nu} X^\mu Y^\nu = n_\mu (PX)^\nu \nabla_\nu (PY)^\mu \quad (\text{B.2.9})$$

for $X, Y \in \Gamma(T \mathcal{M})$ and ∇ the Levi-Civita connection on $(\mathcal{M}, g)$. For $X, Y \in \Gamma(T\Sigma)$, this reduces to

$$K(X, Y) = g(n, \nabla_X Y) = -g(\nabla_X n, Y), \quad (\text{B.2.10})$$

which hence measures the failure of parallel transport of Y along X to remain tangent. The second expression shows that equivalently, it measures the rate of change of the normal vector n along X , as projected onto Y .

Defining the acceleration a as

$$a = \nabla_n n \Leftrightarrow a^\mu = n^\nu \nabla_\nu n^\mu, \quad (\text{B.2.11})$$

we have the following properties:

(i) $K_{\mu\nu} = -P^\lambda{}_\mu \nabla_\lambda n_\nu = -\nabla_\mu n_\nu - n_\mu a_\nu$;

(ii) $K_{\mu\nu} = K_{\nu\mu}$ as a consequence of vanishing torsion;

(iii) $n^\mu K_{\mu\nu} = K_{\mu\nu} n^\nu = 0$.

These follow straightforwardly either from the definition or the alternative characterisation of the extrinsic curvature above.

We denote the trace of the extrinsic curvature by K , and note that

$$K = g^{\mu\nu} K_{\mu\nu} = \gamma^{\mu\nu} K_{\mu\nu} \quad (\text{B.2.12})$$

due to the transverse property (iii).

Finally, we can relate $K_{\mu\nu}$ to the Lie derivative of γ along n as follows:

$$K_{\mu\nu} = -\frac{1}{2}(\mathcal{L}_n \gamma)_{\mu\nu}. \quad (\text{B.2.13})$$

This follows by expanding the right-hand side and making use of

$$\mathcal{L}_n n^\flat = a^\flat. \quad (\text{B.2.14})$$

Note that the Lie derivative of the *vector* n would be zero, $\mathcal{L}_n n = [n, n] = 0$ —in the identity above, we take the Lie derivative of the *covector* $n^\flat$.

B.2.3 Induced Connection and Intrinsic Curvature

B.2.3.1 Induced Connection

Given the Levi-Civita connection ∇ on $(\mathcal{M}, g)$, we define the *induced/three-dimensional/spatial covariant derivative* $\bar{\nabla}$ on the foliation $\Sigma = \{\Sigma_t\}$ as

$$\bar{\nabla}T = P(\nabla T) \Leftrightarrow (\bar{\nabla}_\lambda T)^{\mu\dots}_{\nu\dots} = P^\alpha_\lambda P^\mu_{\beta\dots} P^\gamma_{\nu\dots} \nabla_\alpha T^{\beta\dots}_{\gamma\dots} \quad (\text{B.2.15})$$

for any $T \in \Gamma(T^{(r,s)}\Sigma)$. Note that T must be tangent to Σ —i.e. have no normal components—for this to define a connection on $T\Sigma$. The direction X of the derivative does not necessarily have to be in $T\Sigma$, but if it is, it holds that

$$\bar{\nabla}_X T = P(\nabla_X T), \quad \text{if } X \in \Gamma(T\Sigma). \quad (\text{B.2.16})$$

Otherwise, for general $X \in \Gamma(T\mathcal{M})$, we have

$$\bar{\nabla}_X T = P(\nabla_{PX} T). \quad (\text{B.2.17})$$

The fact that $\bar{\nabla}$ defines a connection on $T\Sigma$ is clear from the fact that ∇ is a connection, and that P is $C^\infty(\mathcal{M})$ -linear. In particular, we have the Leibniz rule

$$\begin{aligned} \bar{\nabla}_X(T \otimes S) &= P(\nabla_{PX}(T \otimes S)) = P(\nabla_{PX}T) \otimes PS + PT \otimes P(\nabla_{PX}S) \\ &= (\bar{\nabla}_X T) \otimes S + T \otimes (\bar{\nabla}_X S). \end{aligned} \quad (\text{B.2.18})$$

Hence, for example, on a 1-form $\eta \in \Gamma(T^*\Sigma)$,

$$(\bar{\nabla}_X \eta)(Y) = X(\eta(Y)) - \eta(\bar{\nabla}_X Y). \quad (\text{B.2.19})$$

Further, $\bar{\nabla}$ is torsion-free and metric-compatible with γ :

- Torsion: For any $X, Y \in \Gamma(T\Sigma)$, we have

$$\begin{aligned} \bar{T}(X, Y) &= \bar{\nabla}_X Y - \bar{\nabla}_Y X - \underbrace{[X, Y]}_{\in \Gamma(T\Sigma)} = P(\nabla_X Y - \nabla_Y X - [X, Y]) \\ &= PT(X, Y) = 0 \end{aligned} \quad (\text{B.2.20})$$

- Metricity: For any $X, Y, Z \in \Gamma(T\Sigma)$,

$$\begin{aligned} X(\gamma(Y, Z)) &= X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \\ &= g(P\nabla_X Y, Z) + g(Y, P\nabla_X Z) = \gamma(\bar{\nabla}_X Y, Z) + \gamma(Y, \bar{\nabla}_X Z). \end{aligned} \quad (\text{B.2.21})$$

B.2.3.2 Intrinsic Curvature

Since $\bar{\nabla}$ defines a connection on $T\Sigma$, and by that on each of the leaves of the foliation, it comes with an associated Riemannian curvature tensor, defined by

$$\bar{R}(X, Y)Z = \bar{\nabla}_X \bar{\nabla}_Y Z - \bar{\nabla}_Y \bar{\nabla}_X Z - \bar{\nabla}_{[X, Y]}Z \quad (\text{B.2.22})$$

In components, this reads

$$(\bar{R}(X, Y)Z)^\mu = \bar{R}^\mu_{\nu\rho\sigma} Z^\nu X^\rho Y^\sigma. \quad (\text{B.2.23})$$

We further have an associated Ricci tensor and scalar,

$$\bar{R}_{\mu\nu} = \bar{R}^\lambda_{\mu\lambda\nu}, \quad \bar{R} = g^{\mu\nu} \bar{R}_{\mu\nu} = \gamma^{\mu\nu} \bar{R}_{\mu\nu}, \quad (\text{B.2.24})$$

where the last equality holds because $\bar{R}^\mu_{\nu\rho\sigma}$ vanishes when contracted with the foliation normal n on any index.

Since $\bar{\nabla}$ is the Levi-Civita connection with respect to γ , $\bar{R}_{\mu\nu\rho\sigma}$ enjoys the same symmetries as the ambient Riemann tensor. Further, the Ricci identity

$$(\bar{\nabla}_\mu \bar{\nabla}_\nu - \bar{\nabla}_\nu \bar{\nabla}_\mu) V^\lambda = \bar{R}^\lambda_{\rho\mu\nu} V^\rho \quad (\text{B.2.25})$$

holds.

B.2.4 Gauss-Codazzi-Ricci Equations

Theorem B.1 (*Gauss Equation*) With the definitions given in the above, the *Gauss equation*

$$(PR)^\lambda_{\rho\mu\nu} = \bar{R}^\lambda_{\rho\mu\nu} + K^\lambda_{\mu} K_{\rho\nu} - K^\lambda_{\nu} K_{\rho\mu} \quad (\text{B.2.26})$$

holds. Contracting over $\lambda\mu$ leads to (after renaming indices)

$$(PR)_{\mu\nu} + R_{\lambda\mu\rho\nu} n^\lambda n^\rho = \bar{R}_{\mu\nu} + K K_{\mu\nu} - K^\lambda_{\mu} K_{\nu\lambda}. \quad (\text{B.2.27})$$

Tracing with respect to $\gamma^{\mu\nu}$ (or equivalently, $g^{\mu\nu}$) leads to

$$R + 2R_{\mu\nu} n^\mu n^\nu = \bar{R} + K^2 - K_{\mu\nu} K^{\mu\nu}. \quad (\text{B.2.28})$$

Proof (Hint) To show this, first compute $\bar{\nabla}_\mu \bar{\nabla}_\nu X^\sigma$ for $X \in \Gamma(T\Sigma)$. Use $K_{\mu\nu} = -P^\lambda_{\mu} \nabla_\lambda n_\nu$ to trade derivatives of n for the extrinsic curvature. $\square$

Theorem B.2 (*Codazzi Equation*) Under the same assumptions, the *Codazzi equation*

$$P(R_{\rho\lambda\mu\nu} n^\rho) = \bar{\nabla}_\mu K_{\nu\lambda} - \bar{\nabla}_\nu K_{\mu\lambda} \quad (\text{B.2.29})$$

holds. Tracing over $\lambda\mu$ with respect to $\gamma^{\lambda\mu}$ (or equivalently, $g^{\lambda\mu}$) leads to

$$P^\mu_{\lambda} n^\nu R_{\mu\nu} = \bar{\nabla}_\lambda K - \bar{\nabla}_\mu K^\mu_{\lambda} \quad (\text{B.2.30})$$

Lemma B.3 With the setup from the above, we have:

(i) The following Lie derivatives of the projector/induced metric:

$$\begin{aligned} \mathcal{L}_n \gamma^{\mu\nu} &= 2K^{\mu\nu} + n^\mu a^\nu + a^\mu n^\nu, \\ \mathcal{L}_n P^\mu_{\nu} &= n^\mu a_\nu, \\ \mathcal{L}_n \gamma_{\mu\nu} &= -2K_{\mu\nu} \end{aligned} \quad (\text{B.2.31})$$

(ii) If $T \in \Gamma(T^{(0,s)}\Sigma)$, that is, if T is a foliation-tangent covariant tensor, then so is $\mathcal{L}_n T$. That is,

$$\mathcal{L}_n T = P(\mathcal{L}_n T), \quad T \in \Gamma(T^{(0,s)}\Sigma). \quad (\text{B.2.32})$$

(iii) The acceleration covector a_μ and the lapse α satisfy the identity

$$a_\mu = P^\lambda_{\mu} \nabla_\lambda \log \alpha = \bar{\nabla}_\mu \log \alpha. \quad (\text{B.2.33})$$

Theorem B.4 (*Ricci Equation*) Again with the same setup, it holds the *Ricci equation*

$$P^\alpha_{\mu} n^\rho P^\beta_{\nu} n^\sigma R_{\alpha\rho\beta\sigma} = \mathcal{L}_n K_{\mu\nu} + K_{\mu\lambda} K^\lambda_{\nu} + \frac{1}{\alpha} \bar{\nabla}_\mu \bar{\nabla}_\nu \alpha \quad (\text{B.2.34})$$

Note that on the left-hand side, the projections of the α and β indices are superfluous, since the $n \otimes n^b$ -part of P vanishes when contracted due to the symmetries of the Riemann tensor. That is, we may write the Ricci equation more compactly as

$$R_{\mu\rho\nu\sigma}n^\rho n^\sigma = \mathcal{L}_n K_{\mu\nu} + K_{\mu\lambda}K^\lambda_\nu + \frac{1}{\alpha}\bar{\nabla}_\mu\bar{\nabla}_\nu\alpha. \quad (\text{B.2.35})$$

Since both sides yield zero when contracted with n , the traces with respect to $g^{\mu\nu}$ and $\gamma^{\mu\nu}$ are identical and read

$$R_{\mu\nu}n^\mu n^\nu = \mathcal{L}_n K - K_{\mu\nu}K^{\mu\nu} + \frac{1}{\alpha}\bar{\nabla}^\mu\bar{\nabla}_\mu\alpha. \quad (\text{B.2.36})$$

With the triad of the Gauss-Codazzi-Ricci equations, we can now isolate the final projection of the Ricci tensor, the purely spatial one, and find an expression for the full Ricci scalar in terms of the intrinsic and extrinsic quantities.

Theorem B.5 The above results combine into

$$\begin{aligned} (PR)_{\mu\nu} &= -\mathcal{L}_n K_{\mu\nu} + K K_{\mu\nu} - 2K_{\mu\lambda}K^\lambda_\nu + \bar{R}_{\mu\nu} - \frac{1}{\alpha}\bar{\nabla}_\mu\bar{\nabla}_\nu\alpha, \\ R &= -2\mathcal{L}_n K + K^2 + K_{\mu\nu}K^{\mu\nu} + \bar{R} - \frac{2}{\alpha}\bar{\nabla}^\mu\bar{\nabla}_\mu\alpha \end{aligned} \quad (\text{B.2.37})$$

B.2.5 The Einstein Equations in the 3 + 1-Formalism

We work with the Einstein equations in the convention where

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}. \quad (\text{B.2.38})$$

Taking the trace of this equation with respect to $g^{\mu\nu}$, we obtain

$$\frac{1}{2}R = 2\Lambda - \frac{1}{2}8\pi T, \quad (\text{B.2.39})$$

with $T = g^{\mu\nu}T_{\mu\nu}$ the trace of the energy-momentum tensor. Inserting this into the Einstein equations yields their trace-reversed counterpart,

$$R_{\mu\nu} = 8\pi\left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}\right) + \Lambda g_{\mu\nu}. \quad (\text{B.2.40})$$

Note that a second trace-reversal brings us back to the original form, implying that the Einstein equations and their trace-reverse are equivalent.

We define the energy density ρ , the energy current j^μ , and the stress tensor $S_{\mu\nu}$ as measured by Eulerian observers traveling along the integral lines of n by

$$\begin{aligned} \rho &= n^\mu n^\nu T_{\mu\nu}, \\ j_\mu &= -P^\alpha_\mu n^\nu T_{\alpha\nu} \\ S_{\mu\nu} &= P^\alpha_\mu P^\beta_\nu T_{\alpha\beta} \end{aligned} \quad (\text{B.2.41})$$

These are simply the time-time, time-space and space-space projections of the energy-momentum tensor, and allow us to decompose it as

$$T_{\mu\nu} = \rho n_\mu n_\nu + j_\mu n_\nu + j_\nu n_\mu + S_{\mu\nu}. \quad (\text{B.2.42})$$

Its trace T is given by

$$T = S - \rho \quad (\text{B.2.43})$$

where $S = g^{\mu\nu} S_{\mu\nu} = \gamma^{\mu\nu} S_{\mu\nu}$.

Together with the results from the previous section, these definitions now allow us to cast the Einstein equations into a first-order system for a second-order in time evolution of the spatial metric $\gamma_{\mu\nu}$, with the extrinsic curvature $K_{\mu\nu}$ serving as an auxiliary variable:

Theorem B.6 The Einstein equations written in terms of $\alpha, n, \gamma, K, \rho, j$ and S read

$$\begin{aligned} \mathcal{H} &:= \bar{R} + K^2 - K_{\mu\nu} K^{\mu\nu} - 2\Lambda - 16\pi\rho = 0, \\ \mathcal{M}_\mu &:= \bar{\nabla}_\mu K - \bar{\nabla}_\nu K^\nu_\mu + 8\pi j_\mu = 0, \\ \mathcal{L}_n \gamma_{\mu\nu} &= -2K_{\mu\nu}, \\ \mathcal{L}_n K_{\mu\nu} &= K K_{\mu\nu} - 2K_{\mu\lambda} K^\lambda_\nu + \bar{R}_{\mu\nu} - \frac{1}{\alpha} \bar{\nabla}_\mu \bar{\nabla}_\nu \alpha \\ &\quad - \Lambda \gamma_{\mu\nu} - 8\pi \left(\left(S_{\mu\nu} - \frac{1}{2} S \gamma_{\mu\nu} \right) + \frac{1}{2} \rho \gamma_{\mu\nu} \right). \end{aligned} \quad (\text{B.2.44})$$

The first two equations are derived by taking the normal-normal and normal-tangential projections of the original Einstein equations and subsequently identifying the definitions of ρ and j as well as applying the Gauss and Codazzi equations, respectively. These two equations involve no time derivatives, and are hence a set of four constraints.

The remaining two equations are first order in time (by the presence of the $\mathcal{L}_n$ normal derivatives), and hence dynamical. The first of the two is simply a consequence of the definition of the extrinsic curvature. The latter is the result of projecting the trace-reversed Einstein equations onto $T\Sigma$ by contracting with P on both indices, and subsequently applying the first equation in (B.2.37) to re-express the projected Ricci tensor $(PR)_{\mu\nu}$.

B.2.6 Adapted Coordinates

B.2.6.1 The Metric and Bases

We can amend the time function $t : \mathcal{M} \rightarrow \mathbb{R}$ (for which $dt \neq 0$) which defines the foliation Σ_t to a coordinate system *adapted to* Σ_t by introducing three additional functions $x^i : \mathcal{M} \rightarrow \mathbb{R}$ such that $dx^i \neq 0$ and $x^i|_{\Sigma_t}$ label points uniquely on any leaf Σ_t . This defines a basis $\{\partial_t, \partial_i\}$ of $T\mathcal{M}$ and an associated dual basis $\{dt, dx^i\}$ of $T^*\mathcal{M}$.

A natural question is to ask how the two time directions that we now have are related—we have the (coordinate-independent) timelike normal vector n , and additionally, the coordinate time direction ∂_t . To this end, we introduce the shift vector β , defined by

$$\beta := \partial_t - \alpha n \quad \Leftrightarrow \quad \partial_t = \alpha n + \beta \quad \Leftrightarrow \quad n = \frac{1}{\alpha}(\partial_t - \beta). \quad (\text{B.2.45})$$

This vector is tangent to the foliation, since

$$dt(\beta) = dt(\partial_t) - \underbrace{\alpha dt(n)}_{=n^\flat} = 1 - g(n, n) = 0. \quad (\text{B.2.46})$$

Hence, it quantifies the failure of ∂_t to be normal to the foliation; that is, how much the coordinate lines of t *shift* with respect to the geometric normal direction specified by n .

With the lapse α , the shift β , and the induced metric γ , we now have all the objects we need for the 3+1-decomposition of the metric tensor g . Its components in the basis $\{\partial_t, \partial_i\}$ read

$$\begin{aligned}
g_{ij} &= g(\partial_i, \partial_j) = \gamma(\partial_i, \partial_j) =: \gamma_{ij} \\
g_{ti} &= g(\partial_t, \partial_i) = \gamma_{ij} \beta^j =: \beta_i, \\
g_{tt} &= g(\partial_t, \partial_t) = -\alpha^2 + \beta_i \beta^i.
\end{aligned} \tag{B.2.47}$$

In matrix form, this reads

$$[g_{\mu\nu}]_{\mu, \nu \in \{t, i\}} = \begin{pmatrix} -\alpha^2 + \beta_k \beta^k & \beta_j \\ \beta_i & \gamma_{ij} \end{pmatrix}, \tag{B.2.48}$$

and tensorially, we may write

$$\begin{aligned}
g &= g_{\mu\nu} dx^\mu \otimes dx^\nu \\
&= -\alpha^2 dt \otimes dt + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt).
\end{aligned} \tag{B.2.49}$$

The corresponding inverse metric has the components

$$g^{tt} = -\frac{1}{\alpha^2}, \quad g^{ti} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i \beta^j}{\alpha^2}, \tag{B.2.50}$$

which takes the matrix form

$$[g^{\mu\nu}]_{\mu, \nu \in \{t, i\}} = \begin{pmatrix} -1/\alpha^2 & \beta^j/\alpha^2 \\ \beta^i/\alpha^2 & \gamma^{ij} - \beta^i \beta^j/\alpha^2 \end{pmatrix}. \tag{B.2.51}$$

Here, γ^{ij} denotes the matrix inverse of γ_{ij} , characterised uniquely by the condition

$$\gamma^{ij} \gamma_{jk} = \delta^i_k. \tag{B.2.52}$$

Tensorially, the inverse metric can be expressed compactly as

$$g^{-1} = -n \otimes n + \gamma^{ij} \partial_i \otimes \partial_j. \tag{B.2.53}$$

As we have seen in the previous sections, we often care about normal projections of tensors, not their coordinate time components. Because of this, another useful basis is $\{n, \partial_i\}$, alongside its dual $\{-n^\flat, dx^i\} = \{\alpha dt, dx^i\}$, with indices $\perp, i$. Generically, this is *not* a coordinate basis. However, it does give the metric a nice block-diagonal structure:

$$g_{\perp\perp} = g(n, n) = -1, \quad g_{\perp i} = g(n, \partial_i) = 0, \quad g_{ij} = g(\partial_i, \partial_j) = \gamma_{ij}, \tag{B.2.54}$$

and

$$g^{\perp\perp} = -1, \quad g^{\perp i} = 0, \quad g^{ij} = \gamma^{ij}. \tag{B.2.55}$$

For example, in terms of this basis, the Hamiltonian and momentum constraints are nothing but the Einstein equations

$$\mathcal{H} = G_{\perp\perp} - 8\pi T_{\perp\perp} = 0, \quad \mathcal{M}_i = G_{\perp i} - 8\pi T_{\perp i}. \tag{B.2.56}$$

In this basis, foliation-tangent tensors, i.e. objects $T \in \Gamma(T^{(r,s)}\Sigma)$, only have components where all indices are spatial. That is, a component vanishes if any of its indices is $\perp$.

Back in the coordinate basis $\{\partial_t, \partial_i\}$, the same is true for upstairs indices; for example, for a vector $X \in \Gamma(T\Sigma)$, we have

$$X^t = dt(X) = -\frac{1}{\alpha} n^\flat(X) = 0, \tag{B.2.57}$$

since $X \perp n$. For downstairs indices, this is generally not true, as, for instance,

$$X_t = g_{t\mu} X^\mu = g_{ti} X^i = \beta_i X^i. \quad (\text{B.2.58})$$

However, the X_t -component is not independent of the other three, $X_i = \gamma_{ij} X^j$. Because of this, it is irrelevant, and hence, the full information about any foliation-tangent tensor $T \in \Gamma(T^{(r,s)}\Sigma)$ (such as $\gamma_{\mu\nu}$, $K_{\mu\nu}$, $\bar{R}_{\mu\nu\rho\sigma}$, $S_{\mu\nu}$ etc.) is stored in its purely spatial components. We also don't need the downstairs-, t components for contractions; it will always be contracted with an upstairs- t component, which is zero. Concretely, this implies that for example,

$$K_{\mu\nu} K^{\mu\nu} = K_{ij} K^{ij} \quad \text{and} \quad \bar{R}_{ij} = \bar{R}^\mu_{i\mu j} = \bar{R}^k_{ikj}. \quad (\text{B.2.59})$$

B.2.6.2 Extrinsic and Intrinsic Curvature

Lemma B.7 For any vector field N that is normal to $T\Sigma$ (not necessarily the unit-normal n), and a contravariant tangential tensor $T \in \Gamma(T^{(0,s)}\Sigma)$, we have

$$\mathcal{L}_{fN} T = f \mathcal{L}_N T. \quad (\text{B.2.60})$$

Note that this is not necessarily true if T is not foliation-tangential or has upstairs indices.

This lemma might seem kind of useless, but it allows us to write down an explicit expression for the extrinsic curvature in an adapted coordinate basis:

$$K_{\mu\nu} = -\frac{1}{2} \mathcal{L}_n \gamma_{\mu\nu} = -\frac{1}{2} \mathcal{L}_{\frac{1}{\alpha}(\partial_t - \beta)} \gamma_{\mu\nu} = -\frac{1}{2\alpha} (\partial_t \gamma_{\mu\nu} - \mathcal{L}_\beta \gamma_{\mu\nu}) \quad (\text{B.2.61})$$

Since the extrinsic curvature itself is also a foliation-tangent tensor, an analogous result holds. Writing directly in the adapted basis (and dropping any t -components), we obtain

$$\begin{aligned} K_{ij} &= -\frac{1}{2\alpha} (\partial_t \gamma_{ij} - \mathcal{L}_\beta \gamma_{ij}), \\ \mathcal{L}_n K_{ij} &= \frac{1}{\alpha} (\partial_t K_{ij} - \mathcal{L}_\beta K_{ij}) \end{aligned} \quad (\text{B.2.62})$$

Writing out the Lie derivatives along β , we obtain the following relationships:

$$\begin{aligned} \partial_t \gamma_{ij} &= \beta^k \partial_k \gamma_{ij} + 2\gamma_{k(i} \partial_{j)} \beta^k - 2\alpha K_{ij}, \\ \partial_t K_{ij} &= \beta^k \partial_k K_{ij} + 2K_{k(i} \partial_{j)} \beta^k + \alpha \mathcal{L}_n K_{ij}. \end{aligned} \quad (\text{B.2.63})$$

These will be useful to recast the 3+1 Einstein equations we derived in the previous section into adapted coordinates, in a form where they clearly describe the first-order time evolution of (γ_{ij}, K_{ij}) .

In the 3+1 Einstein equations, beyond the extrinsic curvature, the intrinsic connection $\bar{\nabla}$ as well as its associated curvature $\bar{R}$ appears. We therefore also need to write these in terms of the adapted coordinate basis. Since $\bar{\nabla}$ is the Levi-Civita connection on any of the leaves with respect to the induced metric γ , its components in adapted coordinates read

$$\bar{\Gamma}^k_{ij} = \frac{1}{2} \gamma^{k\ell} (\partial_i \gamma_{j\ell} + \partial_j \gamma_{i\ell} - \partial_\ell \gamma_{ij}) \quad (\text{B.2.64})$$

Correspondingly, the associated Riemann curvature has the components

$$\bar{R}^k_{\ell ij} = \partial_i \bar{\Gamma}^k_{\ell j} - \partial_j \bar{\Gamma}^k_{\ell i} + \bar{\Gamma}^k_{ni} \bar{\Gamma}^n_{\ell j} - \bar{\Gamma}^k_{nj} \bar{\Gamma}^n_{\ell i} \quad (\text{B.2.65})$$

B.2.6.3 The Einstein Equations in 3 + 1-Adapted Coordinates

The equations (B.2.65) directly imply that the Einstein equations in adapted coordinates reads

$$\mathcal{H} = \bar{R} + K^2 - K_{ij}K^{ij} - 2\Lambda - 16\pi\rho = 0,$$

$$\mathcal{M}_i = \bar{\nabla}_i K - \bar{\nabla}_j K^j_i + 8\pi j_i$$

$$\partial_t \gamma_{ij} = \beta^k \partial_k \gamma_{ij} + 2\gamma_{k(i} \partial_{j)} \beta^k - 2\alpha K_{ij}, \quad (\text{B.2.66})$$

$$\begin{aligned} \partial_t K_{ij} = & \beta^k \partial_k K_{ij} + 2K_{k(i} \partial_{j)} \beta^k - \bar{\nabla}_i \bar{\nabla}_j \alpha + \alpha (K K_{ij} - 2K_{ik} K^k_j + \bar{R}_{ij}) \\ & - \alpha \Lambda \gamma_{ij} - 8\pi \alpha \left(S_{ij} - \frac{1}{2} (S - \rho) \gamma_{ij} \right) \end{aligned}$$

The conservation equation $\nabla_\mu T^{\mu\nu} = 0$ further implies equations of motion for the energy density ρ and j_i . To derive them, we must first relate ρ and j_i to components of the energy-momentum tensor in adapted coordinates. To this end, we should recall that in adapted coordinates,

$$\begin{aligned} n^t &= \frac{1}{\alpha}, & n^i &= -\frac{\beta^i}{\alpha}, \\ n_t &= -\alpha, & n_i &= 0. \end{aligned} \quad (\text{B.2.67})$$

Consequently, the projector P^μ_ν has the components

$$\begin{aligned} P^t_t &= 0, & P^t_i &= 0, \\ P^i_t &= \beta^i, & P^i_j &= \delta^i_j, \end{aligned} \quad (\text{B.2.68})$$

which may be summarised as $P^t_\mu = 0$ and $P^i_\mu = \delta^i_\mu + \beta^i \delta^t_\mu$. With this projector and the components of n , we can directly evaluate

$$\begin{aligned} \rho &= T^{\mu\nu} n_\mu n_\nu = \alpha^2 T^{tt}, \\ j^i &= -P^i_\mu T^{\mu\nu} n_\nu = \alpha (\beta^i T^{tt} + T^{it}), \\ S^{ij} &= P^i_\mu P^j_\nu T^{\mu\nu} = T^{ij} + 2\beta^{(i} T^{j)t} + \beta^i \beta^j T^{tt}. \end{aligned} \quad (\text{B.2.69})$$

Here, we recall that $j^t = 0$, $S^{t\mu} = 0$, and that hence, we may raise and lower indices with γ_{ij} . Note that j_t and $S_{t\mu}$ are not necessarily zero, but by the reasoning given earlier, these components are not independent of the purely spatial ones and hence carry no additional information.

The above puts ρ, j^i and S^{ij} in terms of the components of $T^{\mu\nu}$. To evaluate projections of $\nabla_\mu T^{\mu\nu} = 0$, however, we need to invert this relationship, and express $T^{\mu\nu}$ in terms of ρ, j^i and S^{ij} . Carrying out this inversion leads to

$$T^{tt} = \frac{1}{\alpha^2} \rho, \quad T^{it} = \frac{1}{\alpha} j^i - \frac{\beta^i}{\alpha^2} \rho, \quad T^{ij} = S^{ij} - \frac{2}{\alpha} \beta^{(i} j^{j)} + \frac{\beta^i \beta^j}{\alpha^2} \rho. \quad (\text{B.2.70})$$

Note that this is simply saying that

$$T = \rho n \otimes n + j \otimes n + n \otimes j + S. \quad (\text{B.2.71})$$

The normal projection $(\nabla_\mu T^{\mu\nu}) n_\nu = 0$ then implies the equation of motion for ρ , reading

$$\partial_t \rho = \beta^k \partial_k \rho - 2j^k \partial_k \alpha + \alpha (K \rho + K_{ij} S^{ij} - \bar{\nabla}_i j^i). \quad (\text{B.2.72})$$

This is derived by first integrating by parts, $(\nabla_\mu T^{\mu\nu}) n_\nu = \nabla_\mu (T^{\mu\nu} n_\nu) - T^{\mu\nu} \nabla_\mu n_\nu$, and then applying identities from the above to replace derivatives of n with the extrinsic curvature. The standard formula $\nabla_\mu X^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} X^\mu)$, together with $\sqrt{g} = \alpha \sqrt{\gamma}$, then allows the deduction of the result.

A similar equation of motion can be derived for $\partial_t j_i$ by projecting $\nabla_\mu T^{\mu\nu} = 0$ onto $T\Sigma$ using P , but I will not do this here.

B.3 Metric Derivative and Curvature Identities

B.3.1 Jacobi Formula

For the variation δ of a matrix M , we have the *Jacobi formula*

$$\delta \det M = \det M \cdot \text{tr}(M^{-1} \delta M). \quad (\text{B.3.1})$$

Using $g = \det[g_{\mu\nu}]$, this implies that

$$\delta g = g g^{\mu\nu} \delta g_{\mu\nu} = -g g_{\mu\nu} \delta g^{\mu\nu}. \quad (\text{B.3.2})$$

For a derivative variation, $\delta = \partial_\lambda$, this can be expressed in terms of Christoffel symbols as

$$\partial_\lambda g = g g^{\mu\nu} \partial_\lambda g_{\mu\nu} = 2g \Gamma_{\mu\lambda}^\mu. \quad (\text{B.3.3})$$

This expression further implies expressions for the divergence of $X \in \Gamma(T\mathcal{M})$ and the wave operator on $f \in C^\infty(\mathcal{M})$, reading

$$\begin{aligned} \text{div } X &= \nabla_\mu X^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} X^\mu) \\ &\text{and} \end{aligned} \quad (\text{B.3.4})$$

$$\square_g f = g^{\mu\nu} \nabla_\mu \nabla_\nu f = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} g^{\mu\nu} \partial_\nu f),$$

respectively.

B.3.2 Connection and Curvature of Conformal Metrics

In the following, let $g_{\mu\nu}$ and $\tilde{g}_{\mu\nu}$ be two metrics on a d -dimensional manifold $\mathcal{M}$, related by

$$\tilde{g}_{\mu\nu} = e^{4\phi} g_{\mu\nu}. \quad (\text{B.3.5})$$

Trivially, their inverses and determinants are then related by

$$\tilde{g}^{\mu\nu} = e^{-4\phi} g^{\mu\nu} \quad \text{and} \quad \tilde{g} = e^{4d\phi} g. \quad (\text{B.3.6})$$

Denoting their Christoffel symbols by Γ and $\tilde{\Gamma}$, respectively, we have the identity

$$\tilde{\Gamma}_{\mu\nu}^\lambda = \Gamma_{\mu\nu}^\lambda + 2(\delta_{\mu}^\lambda \partial_\nu \phi + \delta_{\nu}^\lambda \partial_\mu \phi - g_{\mu\nu} g^{\lambda\rho} \partial_\rho \phi) \quad (\text{B.3.7})$$

or equivalently,

$$\Gamma_{\mu\nu}^\lambda = \tilde{\Gamma}_{\mu\nu}^\lambda - 2(\delta_{\mu}^\lambda \partial_\nu \phi + \delta_{\nu}^\lambda \partial_\mu \phi - g_{\mu\nu} g^{\lambda\rho} \partial_\rho \phi). \quad (\text{B.3.8})$$

We note that the difference between the two connection coefficients,

$$C_{\mu\nu}^\lambda := \tilde{\Gamma}_{\mu\nu}^\lambda - \Gamma_{\mu\nu}^\lambda = 2(\delta_{\mu}^\lambda \tilde{\nabla}_\nu \phi + \delta_{\nu}^\lambda \tilde{\nabla}_\mu \phi - \tilde{g}_{\mu\nu} \tilde{g}^{\lambda\rho} \tilde{\nabla}_\rho \phi) \quad (\text{B.3.9})$$

is a tensor, and independent of whether one writes it in terms of $g_{\mu\nu}$ or $\tilde{g}_{\mu\nu}$ in the last term. This means that the Levi-Civita connection actions on an arbitrary tensor $T \in \Gamma(T^{(r,s)}\mathcal{M})$ are related by

$$\tilde{\nabla}_\lambda T^{\mu\dots}_{\nu\dots} = \nabla_\lambda T^{\mu\dots}_{\nu\dots} + C_{\rho\lambda}^\mu T^{\rho\dots}_{\nu\dots} + \dots - C_{\nu\lambda}^\rho T^{\mu\dots}_{\rho\dots} - \dots \quad (\text{B.3.10})$$

Concretely, using the action on vectors, one can derive that

$$R^\lambda_{\rho\mu\nu} = \tilde{R}^\lambda_{\rho\mu\nu} - \tilde{\nabla}_\mu C^\lambda_{\rho\nu} + \tilde{\nabla}_\nu C^\lambda_{\rho\mu} + C^\lambda_{\sigma\mu} C^\sigma_{\rho\nu} - C^\lambda_{\sigma\nu} C^\sigma_{\rho\mu}. \quad (\text{B.3.11})$$

By contracting over $\lambda\mu$, we find a relationship between the Ricci tensors reading

$$R_{\mu\nu} = \tilde{R}_{\mu\nu} + 2\tilde{g}_{\mu\nu}\tilde{g}^{\rho\sigma}\tilde{\nabla}_\rho\tilde{\nabla}_\sigma\phi + 2(d-2)(\tilde{\nabla}_\mu\tilde{\nabla}_\nu\phi + 2\tilde{\nabla}_\mu\phi\tilde{\nabla}_\nu\phi - 2\tilde{g}_{\mu\nu}\tilde{g}^{\rho\sigma}\tilde{\nabla}_\rho\phi\tilde{\nabla}_\sigma\phi) \quad (\text{B.3.12})$$

For the Ricci scalars—taking note that $\tilde{R} = \tilde{g}^{\mu\nu}\tilde{R}_{\mu\nu}$ and $R = g^{\mu\nu}R_{\mu\nu}$ are traced with respect to their respective metric—we obtain

$$R = e^{4\phi}(\tilde{R} + 4(d-1)\tilde{g}^{\mu\nu}\tilde{\nabla}_\mu\tilde{\nabla}_\nu\phi - 4(d-2)(d-1)\tilde{g}^{\mu\nu}\tilde{\nabla}_\mu\phi\tilde{\nabla}_\nu\phi). \quad (\text{B.3.13})$$

B.3.3 Bochner Formula and the Ricci Tensor

For two scalar fields φ and ψ , the *Bochner formula*

$$\begin{aligned} \frac{1}{2}\square_g(\nabla_\mu\varphi\nabla^\mu\psi) &= \frac{1}{2}(\nabla_\mu\square_g\varphi)\nabla^\mu\psi + \frac{1}{2}\nabla_\mu\varphi(\nabla^\mu\square_g\psi) \\ &\quad + \nabla_\mu\nabla_\nu\varphi\nabla^\mu\nabla^\nu\psi + R^{\mu\nu}\nabla_\mu\varphi\nabla_\nu\psi. \end{aligned} \quad (\text{B.3.14})$$

Using the coordinate functions $\varphi = x^\mu$, $\psi = x^\nu$, it follows that

$$R^{\mu\nu} = \frac{1}{2}g^{\lambda\rho}\partial_\lambda\partial_\rho g^{\mu\nu} - \frac{1}{2}\Gamma^\lambda\partial_\lambda g^{\mu\nu} + \partial^{(\mu}\Gamma^{\nu)} - \Gamma^\mu_{\lambda\rho}\Gamma^{\nu\lambda\rho}, \quad (\text{B.3.15})$$

where $\Gamma^\mu = g^{\lambda\rho}\Gamma^\mu_{\lambda\rho}$. Explicitly lowering the indices leads to

$$\begin{aligned} R_{\mu\nu} &= -\frac{1}{2}g^{\lambda\rho}\partial_\lambda\partial_\rho g_{\mu\nu} + g_{\lambda(\mu}\partial_{\nu)}\Gamma^\lambda \\ &\quad + \frac{1}{2}\Gamma^\lambda\partial_\lambda g_{\mu\nu} + g^{\lambda\rho}g^{\sigma\tau}\partial_\lambda g_{\mu\sigma}\partial_\rho g_{\nu\tau} - \Gamma_{\mu\rho\sigma}\Gamma_\nu^{\rho\sigma}. \end{aligned} \quad (\text{B.3.16})$$

This expression will motivate why in BSSNOK, one promotes a version of Γ^μ to be a dynamical variable; in this case, the only second-order part in $R_{\mu\nu}$ is the well-behaved (in that case, elliptic) expression $g^{\lambda\rho}\partial_\lambda\partial_\rho g_{\mu\nu}$.

For numerical implementations, we should address one issue the above has, though. This issue is that the expression we obtained for the Ricci tensor depends on both ∂g as well as Γ . This increases register pressure in GPU-based implementations, which we would like to avoid; hence, we should make use of the identity

$$\partial_\lambda g_{\mu\nu} = \Gamma_{\mu\nu\lambda} + \Gamma_{\nu\mu\lambda} \quad (\text{B.3.17})$$

to re-express everything in terms of Christoffel symbols. Doing so yields

$$\begin{aligned} R_{\mu\nu} &= -\frac{1}{2}g^{\lambda\rho}\partial_\lambda\partial_\rho g_{\mu\nu} + g_{\lambda(\mu}\partial_{\nu)}\Gamma^\lambda \\ &\quad + \Gamma^\lambda\Gamma_{(\mu\nu)\lambda} + \Gamma_{\lambda\rho\mu}\Gamma^{\lambda\rho}_{\nu} + 2\Gamma_{\lambda\rho(\mu}\Gamma_{\nu)}^{\lambda\rho}, \end{aligned} \quad (\text{B.3.18})$$

which is now written entirely in terms of g , $\partial^2 g$, and the two flavours of Γ .

B.4 BSSNOK

In this section, we take what we have derived so far and build atop it the BSSNOK formalism. This involves

- defining the BSSNOK variables in terms of the ADM variables γ_{ij} and K_{ij} , scaling out the conformal factor to isolate the conformal metric $\tilde{\gamma}_{ij}$ and trace-free extrinsic curvature $\tilde{A}_{ij}$;
- deriving their first time derivative expressions from the known expressions for $\partial_t \gamma_{ij}$ and $\partial_t K_{ij}$ via the ADM evolution equations, as well as re-expressing the Hamiltonian and momentum constraints in terms of the new conformal variables;
- and finally, relating the physical induced connections and curvatures to their conformal counterparts using conformal transformations and the auxiliary connection quantities.

In doing so, we will use many of the identities from the previous section, and establish a number of additional ones to handle the conformal rescaling. Unfortunately, these algebraic definitions are rather lengthy and messy, but they ultimately lead to a strongly hyperbolic formulation of general relativity well-suited to be implemented in high-performance computing (HPC) code.

B.4.1 BSSNOK Reparametrisation and Evolution Equations

Definition B.8 (*BSSN Variables*) The dynamical variables in the BSSN formulation of GR are

- (i) the *conformal factor* ϕ , defined as

$$\phi = -\frac{1}{12} \log \gamma, \quad (\text{B.4.1})$$

- (ii) the *conformal metric* $\tilde{\gamma}_{ij}$, given by

$$\tilde{\gamma}_{ij} = e^{4\phi} \gamma_{ij}, \quad (\text{B.4.2})$$

- (iii) the trace of the extrinsic curvature K , defined by

$$K = \gamma^{ij} K_{ij}, \quad (\text{B.4.3})$$

- (iv) the traceless conformal extrinsic curvature $\tilde{A}_{ij}$, expressed as

$$\tilde{A}_{ij} = e^{4\phi} \left(K_{ij} - \frac{1}{3} K \gamma_{ij} \right) \quad (\text{B.4.4})$$

- (v) and the contracted conformal connection $\tilde{\Gamma}^i$, defined as

$$\tilde{\Gamma}^i = \tilde{\gamma}^{k\ell} \tilde{\Gamma}_{k\ell}^i, \quad \text{with} \quad \tilde{\Gamma}_{k\ell}^i = \frac{1}{2} \tilde{\gamma}^{ij} (\partial_k \tilde{\gamma}_{\ell j} + \partial_\ell \tilde{\gamma}_{kj} - \partial_j \tilde{\gamma}_{k\ell}). \quad (\text{B.4.5})$$

Corollary B.9 These definitions have a number of important immediate corollaries:

- (i) The determinant of the conformal metric $\tilde{\gamma}_{ij}$ is 1, that is,

$$\tilde{\gamma} = e^{3 \cdot 4\phi} \gamma = e^{-\log \gamma} \gamma = 1. \quad (\text{B.4.6})$$

This has a few important consequences; firstly, the trace of its Christoffel symbols vanishes,

$$\tilde{\Gamma}_{ki}^i = \frac{1}{2} \tilde{\gamma}^{ij} \partial_k \tilde{\gamma}_{ij} = \frac{1}{\sqrt{\tilde{\gamma}}} \partial_k \sqrt{\tilde{\gamma}} = \partial_k 1 = 0. \quad (\text{B.4.7})$$

Moreover, its contracted Christoffel symbols $\tilde{\Gamma}^i$ may be written as

$$\tilde{\Gamma}^i = -\partial_k \tilde{\gamma}^{ki} \quad (\text{B.4.8})$$

- (ii) $\tilde{A}_{ij}$ is traceless with respect to both $\tilde{\gamma}^{ij}$ and γ^{ij} , that is,

$$\tilde{\gamma}^{ij} \tilde{A}_{ij} = \gamma^{ij} \tilde{A}_{ij} = 0. \quad (\text{B.4.9})$$

- (iii) The definitions of ϕ , $\tilde{\gamma}_{ij}$, K and $\tilde{A}_{ij}$ can be inverted for γ_{ij} and K_{ij} , with the inversions reading

$$\gamma_{ij} = e^{-4\phi} \tilde{\gamma}_{ij} \quad \text{and} \quad K_{ij} = e^{-4\phi} \left(\tilde{A}_{ij} + \frac{1}{3} K \tilde{\gamma}_{ij} \right) \quad (\text{B.4.10})$$

- (iv) From the point of view of the dynamical evolution equations we are about to derive for $\tilde{\gamma}_{ij}$, $\tilde{A}_{ij}$ and $\tilde{\Gamma}^i$, these are two symmetric 3×3 -matrix- and one 3-vector-valued functions. The above tells us that in addition to these evolution equations, we get additional constraints; algebraically, we must always have

$$\tilde{\gamma} = 1, \quad \tilde{\gamma}^{ij} \tilde{A}_{ij} = 0, \quad (\text{B.4.11})$$

and, in addition, a differential constraint emerges from

$$\mathcal{G}^i := \tilde{\Gamma}^i - \tilde{\gamma}^{k\ell} \tilde{\Gamma}_{k\ell}^i = 0. \quad (\text{B.4.12})$$

Although mathematically speaking, the evolution equations will preserve these, numerically, there is a slight drift away from them. The algebraic constraints are very easy to enforce numerically—during discrete evolution we can simply regularly remap

$$\tilde{\gamma}_{ij} \mapsto \tilde{\gamma}^{-1/3} \tilde{\gamma}_{ij}, \quad \tilde{A}_{ij} \mapsto \tilde{A}_{ij} - \frac{1}{3} \tilde{\gamma}_{ij} \tilde{A}^k_k. \quad (\text{B.4.13})$$

To preserve the differential constraint $\mathcal{G}^i = 0$ even in numerical simulations, we will artificially add it to the right-hand side of the evolution equation for $\tilde{\Gamma}^i$, as

$$\partial_t \tilde{\Gamma}^i = -\sigma \mathcal{G}^i + \dots \quad (\text{B.4.14})$$

for some positive constant $\sigma > 0$. This creates exponential damping/decay of $\mathcal{G}^i$ towards 0, so that numerical error does not diverge.

- (v) The trace of the square of K_{ij} reads

$$K_{ij} K^{ij} = \tilde{A}_{ij} \tilde{A}^{ij} + \frac{1}{3} K^2. \quad (\text{B.4.15})$$

Here, indices of quantities with a tilde are raised and lowered using $\tilde{\gamma}$, whereas for quantities without tilde, γ is used. We will continue to use this convention in the following to omit writing hundreds of instances of the induced and conformal metrics.

Proposition B.10 (*BSSNOK Evolution Equations*) Below are the evolution equations for the BSSNOK variables ϕ , $\tilde{\gamma}_{ij}$, K , $\tilde{A}_{ij}$ and $\tilde{\Gamma}^i$.

$$\begin{aligned}
\mathcal{H} &= \bar{R} + \frac{2}{3}K^2 - \tilde{A}_{ij}\tilde{A}^{ij} - 2\Lambda - 16\pi\rho = 0, \\
\mathcal{M}_i &= \frac{2}{3}\partial_i K - \tilde{\nabla}_k \tilde{A}^k_i + 6\tilde{A}^k_i \partial_k \phi + 8\pi j_i = 0, \\
\partial_t \phi &= \beta^k \partial_k \phi - \frac{1}{6}\partial_k \beta^k + \frac{1}{6}\alpha K, \\
\partial_t \tilde{\gamma}_{ij} &= \beta^k \partial_k \tilde{\gamma}_{ij} + 2\tilde{\gamma}_{k(i} \partial_{j)} \beta^k - \frac{2}{3}\tilde{\gamma}_{ij} \partial_k \beta^k - 2\alpha \tilde{A}_{ij}, \\
\partial_t K &= \beta^k \partial_k K - \gamma^{ij} \bar{\nabla}_i \bar{\nabla}_j \alpha + \alpha \left(\tilde{A}_{ij} \tilde{A}^{ij} + \frac{1}{3}K^2 - \Lambda + 4\pi(S + \rho) \right), \\
\partial_t \tilde{A}_{ij} &= \beta^k \partial_k \tilde{A}_{ij} + 2\tilde{A}_{k(i} \partial_{j)} \beta^k - \frac{2}{3}\tilde{A}_{ij} \partial_k \beta^k + e^{4\phi} [\alpha \bar{R}_{ij} - \bar{\nabla}_i \bar{\nabla}_j \alpha - 8\pi \alpha S_{ij}]^{\text{TF}} \\
&\quad + \alpha (K \tilde{A}_{ij} - 2\tilde{A}_{ik} \tilde{A}^k_j) \\
\partial_t \tilde{\Gamma}^i &= \beta^k \partial_k \tilde{\Gamma}^i - \tilde{\Gamma}^k \partial_k \beta^i + \frac{2}{3}\tilde{\Gamma}^i \partial_m \beta^m + \tilde{\gamma}^{j\ell} \partial_j \partial_\ell \beta^i + \frac{1}{3}\tilde{\gamma}^{ij} \partial_j (\partial_k \beta^k) \\
&\quad - 2\tilde{A}^{ij} \partial_j \alpha + 2\alpha \tilde{\Gamma}^i_{jk} \tilde{A}^{jk} - \frac{4}{3}\alpha \tilde{\gamma}^{ij} \partial_j K - 12\alpha \tilde{A}^{ij} \partial_j \phi - 16\pi \alpha \tilde{\gamma}^{ik} j_k - \sigma \mathcal{G}^i \quad (\text{B.4.16})
\end{aligned}$$

In the above, the $^{\text{TF}}$ exponent refers to the trace-free part of the bracketed expression, i.e.

$$X_{ij}^{\text{TF}} = X_{ij} - \frac{1}{3}\gamma_{ij}\gamma^{kl}X_{kl} = X_{ij} - \frac{1}{3}\tilde{\gamma}_{ij}\tilde{\gamma}^{kl}X_{kl}. \quad (\text{B.4.17})$$

Noteworthily, it does not matter whether trace removal is carried out using γ_{ij} or $\tilde{\gamma}_{ij}$.

Remark B.11

- (i) In the derivation of the right-hand side for $\partial_t K$, $\alpha \mathcal{H} = 0$ was subtracted. This does not affect the validity of the equality on-shell, but changes the principal part of the evolution system. Nonetheless, sometimes it is useful to have the same equation without this subtraction. For such situations, we provide it below:

$$\partial_t K = \beta^k \partial_k K - \gamma^{ij} \bar{\nabla}_i \bar{\nabla}_j \alpha + \alpha(K^2 + \bar{R}) - 3\alpha\Lambda + 4\pi\alpha(S - 3\rho) \quad (\text{B.4.18})$$

One such situation is in the derivation of the right-hand side for $\partial_t \tilde{A}_{ij}$ above, where it turns out to be more convenient to use the unsubtracted version. The reason to subtract the Hamiltonian constraint from this equation is to eliminate the intrinsic Ricci scalar $\bar{R}$. In the standard ADM formulation, $\bar{R}$ introduces second spatial derivatives of the metric into $\partial_t K$, which contributes to weak hyperbolicity and numerical instabilities. Subtracting $\alpha \mathcal{H}$ removes all second derivatives of the metric from the right-hand side of $\partial_t K$, which helps cast the BSSNOK evolution system into a strongly hyperbolic form.

- (ii) This version of the equations is only preliminary as it contains expressions involving $\bar{R}_{ij}$, $\bar{R}$ and $\bar{\nabla}$, which are associated with the non-conformal quantity γ_{ij} which is not available directly to the evolution code. Concretely, the offending terms in the above equations which still contain objects tied to γ_{ij} are $\bar{R}$ in the Hamiltonian constraint, the Laplacian $\bar{\Delta}_\gamma \alpha = \gamma^{ij} \bar{\nabla}_i \bar{\nabla}_j \alpha$ and the intrinsic Ricci tensor $\bar{R}_{ij}$. We will need to re-express these in terms of conformal quantities—this requires the results presented below.
- (iii) Since the evolution equation for $\tilde{\Gamma}^i$ contains an artificially introduced damping term $-\sigma \mathcal{G}^i$ with $\sigma > 0$, we present the full derivation of the equation below. We begin by expanding

$$\begin{aligned}
\partial_t \tilde{\Gamma}^i &= -\partial_t \partial_j \tilde{\gamma}^{ij} = -\partial_j \partial_t \tilde{\gamma}^{ij} = \partial_j (\tilde{\gamma}^{ik} \tilde{\gamma}^{j\ell} \partial_t \tilde{\gamma}_{k\ell}) \\
&= \partial_j \left(\tilde{\gamma}^{ik} \tilde{\gamma}^{j\ell} \left(\beta^m \partial_m \tilde{\gamma}_{k\ell} + \tilde{\gamma}_{mk} \partial_\ell \beta^m + \tilde{\gamma}_{m\ell} \partial_k \beta^m - \frac{2}{3} \tilde{\gamma}_{k\ell} \partial_m \beta^m - 2\alpha \tilde{A}_{k\ell} \right) \right) \\
&= \partial_j \left(-\beta^m \partial_m \tilde{\gamma}^{ij} + \tilde{\gamma}^{j\ell} \partial_\ell \beta^i + \tilde{\gamma}^{i\ell} \partial_\ell \beta^j - \frac{2}{3} \tilde{\gamma}^{ij} \partial_m \beta^m - 2\alpha \tilde{A}^{ij} \right) \\
&= -\cancel{(\partial_j \beta^m) \partial_m \tilde{\gamma}^{ij}} - \beta^m \partial_m \underbrace{\partial_j \tilde{\gamma}^{ij}}_{=-\tilde{\Gamma}^i} + \underbrace{(\partial_j \tilde{\gamma}^{j\ell})}_{=-\tilde{\Gamma}^\ell} \partial_\ell \beta^i + \tilde{\gamma}^{j\ell} \partial_j \partial_\ell \beta^i + \cancel{(\partial_j \tilde{\gamma}^{i\ell}) \partial_\ell \beta^j} + \underbrace{\tilde{\gamma}^{i\ell} \partial_j \partial_\ell \beta^j}_{=-\tilde{\Gamma}^i} \\
&\quad - \frac{2}{3} \underbrace{(\partial_j \tilde{\gamma}^{ij})}_{=-\tilde{\Gamma}^i} \partial_m \beta^m - \frac{2}{3} \tilde{\gamma}^{ij} \partial_j \partial_m \beta^m - 2\tilde{A}^{ij} \partial_j \alpha - 2\alpha \partial_j \tilde{A}^{ij} \\
&= \beta^k \partial_k \tilde{\Gamma}^i - \tilde{\Gamma}^k \partial_k \beta^i + \frac{2}{3} \tilde{\Gamma}^i \partial_m \beta^m + \tilde{\gamma}^{j\ell} \partial_j \partial_\ell \beta^i + \frac{1}{3} \tilde{\gamma}^{ij} \partial_j (\partial_k \beta^k) \\
&\quad - 2\tilde{A}^{ij} \partial_j \alpha - 2\alpha \partial_j \tilde{A}^{ij}
\end{aligned} \tag{B.4.19}$$

The last term is still in an unfortunate form, which we can improve by using the momentum constraint $\mathcal{M}^i = 0$. Concretely, it may be used in re-arranged form to replace $\tilde{\nabla}_j \tilde{A}^{ij}$ as

$$\begin{aligned}
\partial_j \tilde{A}^{ij} &= \tilde{\nabla}_j \tilde{A}^{ij} - \tilde{\Gamma}_{jk}^i \tilde{A}^{jk} - \underbrace{\tilde{\Gamma}_{kj}^j}_{=0} \tilde{A}^{ki} \\
&= -\tilde{\Gamma}_{jk}^i \tilde{A}^{jk} + \frac{2}{3} \tilde{\gamma}^{ij} \partial_j K + 6\tilde{A}^{ij} \partial_j \phi + 8\pi j^i
\end{aligned} \tag{B.4.20}$$

Consequently, the evolution equation for $\tilde{\Gamma}^i$, with the artificial damping term $-\sigma \mathcal{G}^i$ introduced, reads

$$\begin{aligned}
\partial_t \tilde{\Gamma}^i &= \beta^k \partial_k \tilde{\Gamma}^i - \tilde{\Gamma}^k \partial_k \beta^i + \frac{2}{3} \tilde{\Gamma}^i \partial_m \beta^m + \tilde{\gamma}^{j\ell} \partial_j \partial_\ell \beta^i + \frac{1}{3} \tilde{\gamma}^{ij} \partial_j (\partial_k \beta^k) \\
&\quad - 2\tilde{A}^{ij} \partial_j \alpha + 2\alpha \tilde{\Gamma}_{jk}^i \tilde{A}^{jk} - \frac{4}{3} \alpha \tilde{\gamma}^{ij} \partial_j K - 12\alpha \tilde{A}^{ij} \partial_j \phi - 16\pi \alpha j^i - \sigma \mathcal{G}^i
\end{aligned} \tag{B.4.21}$$

- (iv) The keen-eyed reader might have noticed that for any of the BSSN variables, its time derivative contains similarly structured spatial derivative terms of that variable involving the shift β on the right hand side. These combinations are reminiscent of Lie derivatives—and in fact, they are—with a catch. The catch comes from the terms involving the divergence $\partial_i \beta^i$, which in a tensorial Lie derivative do not appear. However, we should recall that ϕ is a function of the *determinant* of the tensor γ_{ij} , and that in the definitions of $\tilde{\gamma}_{ij}$ as well as $\tilde{A}_{ij}$ include factors of

$$e^{4\phi} = \gamma^{-1/3}. \tag{B.4.22}$$

From this, we conclude that ϕ is not a scalar but a scalar *density* of weight $-\frac{1}{6}$, and that $\tilde{\gamma}_{ij}$ and $\tilde{A}_{ij}$ are $(0, 2)$ -tensor *densities* of weight $-\frac{2}{3}$. We recall that for a tensor density $T^{\mu\dots}_{\nu\dots}$ of weight w , its Lie derivative along some vector field X is given by

$$\begin{aligned}
\mathcal{L}_X T^{\mu\dots}_{\nu\dots} &= X^\lambda \partial_\lambda T^{\mu\dots}_{\nu\dots} - (\partial_\lambda X^\mu) T^{\lambda\dots}_{\nu\dots} - \dots \\
&\quad + (\partial_\nu X^\lambda) T^{\mu\dots}_{\lambda\dots} + \dots + w(\partial_\lambda X^\lambda) T^{\mu\dots}_{\nu\dots}
\end{aligned} \tag{B.4.23}$$

A similar expression can be obtained for connections as well. Taking $X = \beta$, $T = \phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}$ and $\tilde{\Gamma}^i$, we identify that the BSSNOK evolution equations in (B.4.16) may be written in the (slightly) more compact form

$$\begin{aligned}
\partial_t \phi &= \mathcal{L}_\beta \phi + \frac{1}{6} \alpha K, \\
\partial_t \tilde{\gamma}_{ij} &= \mathcal{L}_\beta \tilde{\gamma}_{ij} - 2\alpha \tilde{A}_{ij}, \\
\partial_t K &= \mathcal{L}_\beta K - \gamma^{ij} \bar{\nabla}_i \bar{\nabla}_j \alpha + \alpha \left(\tilde{A}_{ij} \tilde{A}^{ij} + \frac{1}{3} K^2 - \Lambda + 4\pi(S + \rho) \right), \\
\partial_t \tilde{A}_{ij} &= \mathcal{L}_\beta \tilde{A}_{ij} + e^{4\phi} [\alpha \bar{R}_{ij} - \bar{\nabla}_i \bar{\nabla}_j \alpha - 8\pi \alpha S_{ij}]^{\text{TF}} + \alpha (K \tilde{A}_{ij} - 2\tilde{A}_{ik} \tilde{A}_j^k), \\
\partial_t \tilde{\Gamma}^i &= \mathcal{L}_\beta \tilde{\Gamma}^i - 2\tilde{A}^{ij} \partial_j \alpha + 2\alpha \tilde{\Gamma}_{jk}^i \tilde{A}^{jk} - \frac{4}{3} \alpha \tilde{\gamma}^{ij} \partial_j K - 12\alpha \tilde{A}^{ij} \partial_j \phi - 16\pi \alpha \tilde{\gamma}^{ik} j_k - \sigma \mathcal{G}^i.
\end{aligned} \tag{B.4.24}$$

Note that the trace of the extrinsic curvature, K , is the only true scalar, so $\mathcal{L}_\beta K = \beta^i \partial_i K$. Moreover, it should be noted that $\mathcal{L}_\beta \tilde{\Gamma}^i$ is *not* to be interpreted as the Lie derivative of a vector field nor vector density, but rather expanded as

$$\mathcal{L}_\beta \tilde{\Gamma}^i = \mathcal{L}_\beta (\tilde{\Gamma}_{jk}^i \tilde{\gamma}^{jk}) = (\mathcal{L}_\beta \tilde{\Gamma}_{jk}^i) \tilde{\gamma}^{jk} + \tilde{\Gamma}_{jk}^i (\mathcal{L}_\beta \tilde{\gamma}^{jk}). \tag{B.4.25}$$

The Lie derivative of the connection coefficients is then evaluated using the analogon of equation (B.4.23) for connections, and gives rise to the terms involving second derivatives of β in the last of equations (B.4.16).

Although this compactification of the BSSN evolution equations does not introduce any benefits to its numerical implementation—we still have to compute all the individual terms the Lie derivatives consist of—writing them down in this form enables a more straightforward interpretation. For any of the $T = \phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}, \tilde{\Gamma}^i$, we have the main Lie-advection piece

$$\partial_t T = \mathcal{L}_\beta T. \tag{B.4.26}$$

In absence of any other terms, this implies that the BSSN variables are Lie-transported along the normal vector field $\alpha n = \partial_t - \beta$ and thus remain invariant under its generated diffeomorphisms. However, there are other terms; these source the Lie transport by introducing curvature, matter and gauge contributions, such as $e^{4\phi} \alpha \bar{R}_{ij}$, $8\pi e^{4\phi} \alpha S_{ij}^{\text{TF}}$ and $\gamma^{ij} \bar{\nabla}_i \bar{\nabla}_j \alpha$, respectively. Put differently, this means that the failure of the BSSN variables to be Lie-transported across the leaves of the foliation is caused by global curvature, the presence of matter, and the choice of the foliation and coordinates on it.

B.4.2 From Bar to Tilde

As alluded to before, the BSSN equations (B.4.16) are preliminary in the sense that they still contain expressions associated with the induced metric γ_{ij} , which is not part of the BSSN variables. Although technically, one can always reconstruct γ_{ij} from ϕ and $\tilde{\gamma}_{ij}$ as $\gamma_{ij} = e^{-4\phi} \tilde{\gamma}_{ij}$, this is computationally inefficient and induces more memory usage and traffic. In this section, we address the offending terms and reexpress them in terms of the BSSN variables $\phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}$ and $\tilde{\Gamma}^i$. Concretely, the terms that appear which we need to reformulate are

$$\bar{R}_{ij}, \quad \bar{R}, \quad \bar{\nabla}_i \bar{\nabla}_j \alpha \quad \text{and} \quad \Delta_\gamma \alpha = \gamma^{ij} \bar{\nabla}_i \bar{\nabla}_j \alpha. \tag{B.4.27}$$

We begin with the last of the four, as it is the simplest. Using the standard formula for the Laplacian, we get

$$\begin{aligned}
\Delta_\gamma \alpha &= \frac{1}{\sqrt{\gamma}} \partial_i (\sqrt{\gamma} \gamma^{ij} \partial_j \alpha) = \frac{1}{\sqrt{e^{-12\phi}}} \partial_i (\sqrt{e^{-12\phi}} e^{4\phi} \tilde{\gamma}^{ij} \partial_j \alpha) \\
&= e^{6\phi} \partial_i (e^{-2\phi} \tilde{\gamma}^{ij} \partial_j \alpha) = e^{4\phi} (\partial_i (\tilde{\gamma}^{ij} \partial_j \alpha) - 2\tilde{\gamma}^{ij} \partial_i \phi \partial_j \alpha) \\
&= e^{4\phi} (\Delta_{\tilde{\gamma}} \alpha - 2\tilde{\gamma}^{ij} \partial_i \phi \partial_j \alpha).
\end{aligned} \tag{B.4.28}$$

In the last step, we made use of the fact that $\tilde{\gamma} = 1$.

Next up, we consider the Hessian of α . For this, we recall the relationship (B.3.10), which allows us to infer

$$\begin{aligned}\bar{\nabla}_i \bar{\nabla}_j \alpha &= \bar{\nabla}_i \partial_j \alpha = \tilde{\nabla}_i \partial_j \alpha + C^k_{ji} \partial_k \alpha \\ &= \tilde{\nabla}_i \tilde{\nabla}_j \alpha + 2(\delta^k_i \partial_j \phi + \delta^k_j \partial_i \phi - \tilde{\gamma}_{ij} \tilde{\gamma}^{k\ell} \partial_\ell \phi) \partial_k \alpha \\ &= \tilde{\nabla}_i \tilde{\nabla}_j \alpha + 4\partial_{(i} \phi \partial_{j)} \alpha - 2\tilde{\gamma}_{ij} \tilde{\gamma}^{k\ell} \partial_k \phi \partial_\ell \alpha.\end{aligned}\tag{B.4.29}$$

Note that since $\tilde{\gamma}^{ij} \tilde{\gamma}_{ij} = 3$, this is consistent with equation (B.4.28).

Now turning to $\bar{R}_{ij}$, we can simply employ our result (B.3.12) for the conformal Ricci tensor, which for $d = 3$ implies

$$\bar{R}_{ij} = \tilde{R}_{ij} + 2(\tilde{\nabla}_i \tilde{\nabla}_j \phi + \tilde{\gamma}_{ij} \tilde{\gamma}^{k\ell} \tilde{\nabla}_k \tilde{\nabla}_\ell \phi) + 4(\partial_i \phi \partial_j \phi - \tilde{\gamma}_{ij} \tilde{\gamma}^{k\ell} \partial_k \phi \partial_\ell \phi).\tag{B.4.30}$$

Here, according to equation (B.3.18), the conformal Ricci tensor $\tilde{R}_{ij}$ is given by

$$\begin{aligned}\tilde{R}_{ij} &= -\frac{1}{2} \tilde{\gamma}^{k\ell} \partial_k \partial_\ell \tilde{\gamma}_{ij} + \tilde{\gamma}_{k(i} \partial_{j)} \tilde{\Gamma}^k \\ &\quad + \tilde{\Gamma}^k \tilde{\Gamma}_{(ij)k} + \tilde{\Gamma}_{k\ell i} \tilde{\Gamma}^{k\ell}_{j} + 2\tilde{\Gamma}_{k\ell(i} \tilde{\Gamma}_{j)}^{k\ell}.\end{aligned}\tag{B.4.31}$$

Here it becomes apparent why introducing $\tilde{\Gamma}^i$ as an additional dynamical variable is a good idea: since it only appears with first derivatives, the principal part of the conformal Ricci tensor is the elliptic operator

$$\tilde{\gamma}^{k\ell} \partial_k \partial_\ell \tilde{\gamma}_{ij}\tag{B.4.32}$$

which is much more well-behaved numerically than the second-order terms introduced by $\partial_i \tilde{\Gamma}^j$ would be. The introduction of $\tilde{\Gamma}^i$ turns exactly these terms into mere first-order contributions, so that they do not mess up the principal symbol.

Lastly, for the Ricci scalar, we may use equation (B.3.13), which for $d = 3$ reads

$$R = e^{4\phi} (\tilde{R} + 8\tilde{\gamma}^{ij} (\tilde{\nabla}_i \tilde{\nabla}_j \phi - \partial_i \phi \partial_j \phi)).\tag{B.4.33}$$

For quicker reference, we provide all these results again below:

$$\begin{aligned}\bar{\nabla}_i \bar{\nabla}_j \alpha &= \tilde{\nabla}_i \tilde{\nabla}_j \alpha + 4\partial_{(i} \phi \partial_{j)} \alpha - 2\tilde{\gamma}_{ij} \tilde{\gamma}^{k\ell} \partial_k \phi \partial_\ell \alpha, \\ \gamma^{ij} \bar{\nabla}_i \bar{\nabla}_j \alpha &= e^{4\phi} (\Delta_{\tilde{\gamma}} \alpha - 2\tilde{\gamma}^{ij} \partial_i \phi \partial_j \alpha), \\ \bar{R}_{ij} &= \tilde{R}_{ij} + 2(\tilde{\nabla}_i \tilde{\nabla}_j \phi + \tilde{\gamma}_{ij} \tilde{\gamma}^{k\ell} \tilde{\nabla}_k \tilde{\nabla}_\ell \phi) + 4(\partial_i \phi \partial_j \phi - \tilde{\gamma}_{ij} \tilde{\gamma}^{k\ell} \partial_k \phi \partial_\ell \phi) \\ \text{with } \tilde{R}_{ij} &= -\frac{1}{2} \tilde{\gamma}^{k\ell} \partial_k \partial_\ell \tilde{\gamma}_{ij} + \tilde{\gamma}_{k(i} \partial_{j)} \tilde{\Gamma}^k \\ &\quad + \tilde{\Gamma}^k \tilde{\Gamma}_{(ij)k} + \tilde{\Gamma}_{k\ell i} \tilde{\Gamma}^{k\ell}_{j} + 2\tilde{\Gamma}_{k\ell(i} \tilde{\Gamma}_{j)}^{k\ell}, \\ \bar{R} &= e^{4\phi} (\tilde{R} + 8\tilde{\gamma}^{ij} (\tilde{\nabla}_i \tilde{\nabla}_j \phi - \partial_i \phi \partial_j \phi)).\end{aligned}\tag{B.4.34}$$

B.4.3 Gauge Dynamics

B.5 Initial Data

B.5.1 York-Lichnerowicz

B.5.2 Conformal Transverse-Traceless (CTT) Decomposition

B.6 Boundary Conditions and Grid Stability

B.6.1 Kreiss-Oliger Dissipation

B.6.2 Sommerfeld Radiative Boundaries

B.7 Wave Extraction & Diagnostics

B.7.1 Weyl Scalar Ψ_4

B.7.2 Constraint Monitoring